

ESTIMATING RUN-TO-RUN COVARIANCE ACROSS LANGUAGE MODEL BENCHMARKS

Anonymous authors

Paper under double-blind review

ABSTRACT

Estimating the run-to-run uncertainty of a benchmark average requires covariance between scores as well as their marginal variances. We study this covariance with SNAP, a cross-half estimator that separates item-general run variation from item-sampling noise under conditional independence assumptions. We analyse 375 DataDecide runs across 125 configurations and ten benchmarks, scoring replicates at a common checkpoint step. Relative to independent benchmark deviations, the estimated standard deviation of the average increases by a factor of 1.244 for per-byte margins and 1.078 for accuracy, with nominal 95% recipe-cluster intervals of $[1.143, 1.338]$ and $[0.993, 1.157]$. Simulated coverage of those intervals lies between 0.933 and 0.953 across twelve populations. On the 123 configurations with an earlier shared checkpoint, scoring that checkpoint moves the estimates from 1.233 to 1.192 and from 1.082 to 1.130, and the accuracy interval then excludes one, although paired intervals for both changes include zero. Across single-benchmark removals, the margin factor ranges from 1.096 to 1.786 and the accuracy factor from 1.014 to 1.558, with BoolQ removal at the top of both ranges, so the full-battery estimates depend on its inclusion. Assigning whole BoolQ passages to halves, using passages recovered from the release files, changes both estimates by less than 0.001 on average across fifty re-splits. Margin covariance concentrates within scoring formats, and gain simulations show sensitivity to score scale. Given held-out marginal variances, a full correlation model predicts the margin aggregate standard deviation better than independence under paired recipe-bootstrap intervals and three treatments of negative variance estimates, but no model separates from independence on accuracy, and an operational comparison with paired intervals finds no model with a detectable advantage over independence. The planned recipe holdout was never implemented, so all empirical analyses are exploratory. We recommend reporting replicate uncertainty and sensitivity to benchmark composition, with the score-scale mechanism still unresolved.

1 INTRODUCTION

A benchmark average can change across training runs even when the evaluation items stay fixed. For an equally weighted average of K benchmarks, independent run deviations give variance $K^{-2} \text{tr } \Sigma_E$, where Σ_E is their cross-benchmark covariance matrix. Covariance adds the off-diagonal terms to that variance. A collection of marginal standard deviations alone doesn't determine the uncertainty of the average. Researchers need paired benchmark scores from replicate runs to estimate that contribution.

We estimate item-general covariance with two disjoint item halves for each benchmark. Within each configuration, we centre both half scores across replicate runs, then take their cross-half products. Under zero-mean item noise and conditional independence across halves, these products estimate run covariance, including its diagonal. We adapt the established repeated-measurement construction to paired benchmark scores from replicate runs. We call its application to benchmark replicates SNAP, Seed Noise Across Phenotypes, and use phenotype to mean the measured score.

Our empirical study uses DataDecide (Magnusson et al., 2025), with 25 data recipes and five model sizes that each have three released replicates. Those 125 configurations supply 375 runs across ten benchmarks. We read released item scores, and the primary analysis requires no model inference.

We compare per-byte log-likelihood margins with accuracy on the same items, because these scales respond differently to changes in confidence.

Training conditions vary across the released runs in this population. Auxiliary replicates use different training budgets below 1B parameters, and 92 configurations have mismatched final checkpoint steps. We compare the largest step shared by all three runs, but this rule doesn’t align their training schedules. So we interpret the full estimate as covariance among the released runs at the selected steps, and report an auxiliary-run contrast that removes the default-versus-auxiliary batch component.

The accuracy estimate is close to independence for the full battery, but that result changes with benchmark composition. BoolQ contributes 84% of the estimated accuracy variance trace, and assigning its passages whole to item halves leaves the estimates within re-split variation. Removing BoolQ increases the accuracy inflation factor from 1.078 to 1.558. On the 123 configurations with an earlier shared checkpoint, scoring that checkpoint moves accuracy inflation from 1.082 to 1.130, with an interval that excludes one, although the paired interval for the change includes zero. Deleting parts of the design moves that conclusion in one direction only, because the accuracy interval stops including one under five of the 25 recipe removals, the 750M band, and WinoGrande, and all put the interval above one, a count a null sweep reaches in 0.018 of replicates. Given this sensitivity, we don’t read a small full-battery estimate as evidence of independence in general, and the design also lacks the power to resolve an effect that size, since a simulation at the fitted accuracy cell excludes one in 0.111 of replicates at our design, against 0.970 on margins.

We define the item-general run covariance that cross-half products estimate, and we state the conditions under which released language model runs satisfy its assumptions, including a passage-aware check on the one benchmark with shared passages. We measure that covariance on DataDecide on two score scales, with recipe-cluster intervals whose simulated coverage lies between 0.933 and 0.953. We then examine batch, gain, and format mechanisms that could produce the observed covariance, and we test covariance models on held-out recipe folds with paired uncertainty. The initial analysis plan specified a separate screening set that we never formed, its date has no independent timestamp, and none of the analyses supports a confirmatory interpretation.

2 ESTIMAND AND ESTIMATOR

2.1 RUNS AND SCORE SCALES

Let c index configurations, r index their R replicate runs, and j index benchmarks. A configuration fixes a data recipe, model size, and nominal hyperparameters. In DataDecide, the seed controls initialisation and data order together, and auxiliary training schedules introduce an additional source of run variation. For item i , define the per-byte margin and corresponding correctness indicator by

$$m_{crji} = \frac{\ell(\text{gold}_i)}{\text{bytes}(\text{gold}_i)} - \max_{v \neq \text{gold}_i} \frac{\ell(v_i)}{\text{bytes}(v_i)}, \quad a_{crji} = \mathbf{1}\{m_{crji} > 0\}. \quad (1)$$

Here ℓ is the summed answer log-likelihood, and v ranges over alternatives. This definition treats a zero margin as incorrect. We average items within each benchmark, with MMLU macro-averaged over its 57 subjects, and denote the resulting margin and accuracy scores by y^M and y^A , calling each benchmark-level score a trait.

For each half $H \in \{A, B\}$, our measurement model is

$$y_{crj}^H = \mu_{cj}^H + E_{crj} + \varepsilon_{crj}^H. \quad (2)$$

A half-specific mean lets the two item sets differ in difficulty. The component E_{crj} is the run deviation shared by the two item populations, and ε_{crj}^H contains the remaining item-dependent variation. We assume independent, identically distributed run effects within a configuration when deriving unbiasedness, and write their covariance as $\Sigma_{E,c}$. We can’t sustain that interpretation in full, because training schedules differ within the actual population.

2.2 CROSS-HALF COVARIANCE

We partition each benchmark’s items into disjoint halves with a frozen random seed, using the same assignment for its replicate runs. We centre each half score across the replicate runs, defining

$d_{crj}^H = y_{crj}^H - R^{-1} \sum_s y_{csj}^H$. If the item errors have conditional mean zero and zero covariance between the two halves across the relevant runs and benchmarks, then

$$\mathbb{E}[d_{crj}^A d_{crk}^B] = (1 - 1/R) \Sigma_{E,c}(j, k). \quad (3)$$

This identity covers diagonal entries, and it doesn't require Gaussian run effects. Disjoint item identifiers alone don't establish the required independence, because passages, duplicate content, and other shared item effects can connect the halves. We check the one benchmark with documented shared passages by assigning whole passages to halves in Section 5.1.

We write the average centred score as $g_{cr}^H = K^{-1} \sum_j d_{crj}^H$ and compute

$$T_c = \frac{1}{R-1} \sum_r g_{cr}^A g_{cr}^B, \quad U_c = \frac{1}{K^2(R-1)} \sum_{j,r} d_{crj}^A d_{crj}^B. \quad (4)$$

The respective expectations of these two statistics are $K^{-2} \mathbf{1}^\top \Sigma_{E,c} \mathbf{1}$ and $K^{-2} \text{tr} \Sigma_{E,c}$. We sum each statistic across configurations before dividing, which gives

$$\hat{\Lambda} = \sqrt{\frac{\sum_c T_c}{\sum_c U_c}}, \quad \Lambda^2 = \frac{\mathbf{1}^\top \Sigma_E \mathbf{1}}{\text{tr} \Sigma_E} = 1 + (K-1) \bar{r}_E, \quad K_{\text{eff}} = \frac{K}{\Lambda^2}, \quad (5)$$

where $\Sigma_E = N^{-1} \sum_c \Sigma_{E,c}$ is the covariance matrix averaged across configurations. We express the off-diagonal contribution through the effective coefficient

$$\bar{r}_E = \frac{\sum_{j \neq k} \Sigma_E(j, k)}{(K-1) \text{tr} \Sigma_E}. \quad (6)$$

This effective covariance coefficient equals the mean off-diagonal correlation when the marginal variances are equal. With unequal variances, it differs from a normalised weighted mean correlation, as Appendix A explains. The effective count compares the battery with independent benchmarks at the same root-mean-square marginal deviation. It isn't a count of independent information sources.

Unbiased estimates of T_c and U_c don't make their ratio or square root unbiased, and either pooled sum can be nonpositive in a finite sample. We use calibration simulations to check finite-sample behaviour in specified populations, where the ratio and square-root transformations can introduce bias.

2.3 RESOLUTION AND INFERENCE

For an infinitesimal additive shift in Gaussian item margins, the ratio of accuracy to margin noise per unit squared sensitivity is

$$\frac{p(1-p)}{\phi(z_0)^2}, \quad z_0 = \Phi^{-1}(1-p). \quad (7)$$

At accuracy $p = 0.35$, this model gives a ratio of about 1.66. Its minimum is $\pi/2$, which means the planned lower threshold of 1.4 couldn't falsify this model. This calculation describes only the assumed location-shift model, and it doesn't show an empirical information advantage for arbitrary margin distributions. The aggregate reliability calculation uses $\rho_g = \Lambda^2/(\Lambda^2 + \bar{v})$, with $\bar{v}$ the item-noise contribution on that scale.

We report nominal 95% wild cluster bootstrap- t intervals over the 25 recipe clusters, using 4,999 Rademacher draws and a linearisation of the squared ratio. Appendix D states the implementation, including its fixed observed denominator during resampling. We also report a configuration percentile bootstrap, which resamples both sums but ignores recipe dependence, so the two procedures make different assumptions, and neither interval width alone identifies a design effect.

Recomputing the seed-label permutation in ten runs of 1,000 permutations each gives margin coverage between 0.915 and 0.942 against the permutation reference value of one, with a mean of 0.931, and accuracy coverage between 0.925 and 0.959 with a mean of 0.944. The margin interval falls below the null in 0.032 to 0.052 of permutations across those ten runs, against a nominal 0.025 on that side. A separate simulation with 2,000 replicates per population and the same 4,999 draws gives

wild bootstrap coverage between 0.928 and 0.954 across twelve populations. Those populations include recipe-shared run effects, heavy-tailed run effects, one recipe with six times the variance, and cross-half error correlation of 0.1. We report the wild interval because the configuration percentile bootstrap drops to 0.872 under recipe-shared effects, and because a recipe percentile bootstrap covers between 0.857 and 0.940. Cross-half error correlation of 0.1 biases the estimate downward by about 0.015, which places residual item dependence on the side of understating inflation. Under accuracy-like item noise, the null population gives a mean estimate of 0.989, a downward bias of 0.011, and the population at 1.078 shows a bias of -0.007 , so finite-sample bias understates rather than manufactures an accuracy excess. Two repeats at 10,000 replicates put wild coverage between 0.933 and 0.953 and between 0.929 and 0.950, so the shortfall isn’t simulation noise. A cluster test-inversion interval that we added later covers between 0.935 and 0.956 in the same populations, and on the observed data it gives $[1.095, 1.323]$ for margins and $[0.995, 1.162]$ for accuracy, so it moves the margin lower endpoint down by 0.048.

3 DATA AND ANALYSIS SCOPE

DataDecide supplies the ten benchmarks ARC-Challenge, ARC-Easy, BoolQ, CommonsenseQA, HellaSwag, MMLU, OpenBookQA, PIQA, Social IQa, and WinoGrande. We analyse 37,682 items per run across 25 recipes at 150M, 300M, 530M, 750M, and 1B parameters. Smaller released models lack the required per-item replicate coverage. Replicate seed labels are $\{2, 14, 15\}$ below 1B and $\{2, 4, 5\}$ at 1B.

We select the largest checkpoint step available for all three runs in each configuration. In 26 configurations, one at 530M and 25 at 750M, the largest final step exceeds the smallest by more than a factor of 1.5, and the shared step then truncates the longer runs. An index of the release’s checkpoint files shows that the selected step averages 87% of the default run’s final step at 530M, with a range of 39% to 89%, and 41% at 750M, with a range of 18% to 44%. At 750M the default runs therefore stop well short of their released final models, although the auxiliary runs sit close to their own final steps. Dropping the 26 severely truncated configurations gives margin inflation of 1.276 with interval $[1.114, 1.420]$, and a configuration’s influence on the estimate shows no detectable rank correlation with its truncation, as Appendix B.1 reports.

The analysis plan assigned eight recipes to screening and seventeen to estimation, but we used all recipes without drawing that partition. As a result, analysis choices used all 125 configurations, and the plan has neither a supplied original file nor an independently corroborated date. We report its thresholds as historical targets because their timing remains unverified. An enumeration over the $\binom{25}{17} = 1,081,575$ possible estimation sets produces margin estimates from 1.037 to 1.317 and accuracy estimates from 0.963 to 1.191. None reaches the planned thresholds of 1.349 and 1.40, but this retrospective calculation can’t restore a holdout or account for choices made after viewing the data.

We retain the full ten-benchmark battery for the primary estimand, including benchmarks with negative estimated diagonal covariance. An unbiased moment estimate can be negative near zero, and deleting a benchmark after observing that sign changes both selection and the estimand. Standardised diagnostics, which use only defined positive-variance traits, carry this selection limitation.

We use PolyPythias (van der Wal et al., 2025) for an exploratory transfer check, scoring nine seeds at each of 70M, 160M, and 410M on 4,755 nested items. Its three configurations can’t support a precise population comparison. We also examine released run-level scores from Heineman et al. (2025), whose single-configuration panels lack item-level outputs for the cross-half correction. A later five-size PolyPythias analysis adds 14M and 31M under a different scoring batch budget.

4 SENSITIVITY TO BATCH AND SCORE SCALE

Default and auxiliary runs come from different batches. We decompose within-configuration contrasts into a default-versus-auxiliary component and an auxiliary-versus-auxiliary component. The latter removes an offset shared by both auxiliary runs, while retaining one contrast per configuration. It doesn’t confirm matching software provenance or remove differences shared with the selected checkpoint schedules. A simulated batch offset provides a separate sensitivity check.

A positive multiplicative gain changes margin magnitudes while preserving their signs, so it leaves threshold accuracy unchanged by construction. We examine this mechanism with a gain covariate s_{cr} , the mean per-byte answer log-likelihood, and with simulations that vary the gain scale and margin distribution. We also fit a leave-one-benchmark-out competence proxy $x_{cr}^{(-j)}$ on the opposite item half, using

$$y_{crj} = \alpha_{cj} + \beta_j x_{cr}^{(-j)} + \gamma_j s_{cr} + u_{crj}. \quad (8)$$

The pooled fit uses one slope per benchmark over 250 run contrasts. These adjustments are descriptive, because the gain covariate shares item noise with the outcomes and the competence proxy uses unstable estimated variance normalisation. The implementation uses a standard-deviation floor of 10^{-6} when variance estimates are negative, making WinoGrande dominate the margin proxy by factors of about 800 to 38,000. Dropping WinoGrande from the proxy moves margin inflation to 0.890, and a recipe cross-fitted proxy with shrunken variances gives 0.731, but simulations with the observed seed covariance and no separate competence variable return 0.889 and 0.745 under the same adjustments, as Appendix C reports. The proxy averages the same seed deviations that it adjusts, so each of these adjustments removes shared covariance by construction, and we give none of them a mechanistic reading.

5 RESULTS

5.1 FULL-BATTERY ESTIMATES AND COMPOSITION

Full-battery inflation is 1.244 on margins and 1.078 on accuracy, as Table 1 reports for both score scales. On margins the raw interval excludes one, but on accuracy it includes it. The accuracy upper endpoint maps to $\bar{r}_E < 0.038$ and $K_{\text{eff}} > 7.47$ under the stated measurement model and fixed battery. Those values are conditional on the measurement model, and the passage-aware split below moves them within re-split variation.

Table 1. Inflation estimates across 125 configurations with nominal 95% recipe-cluster intervals. The auxiliary contrast compares the two auxiliary runs, and the adjustment includes the gain covariate alone. Aggregate standard deviations use margin units for margins and fractions for accuracy.

Score	Estimate	$\hat{\Lambda}$	Interval	K_{eff}	σ_{agg}
Margin	Raw	1.244	[1.143, 1.338]	6.46	0.00572
Margin	Gain adjusted	1.275	[1.111, 1.421]	6.15	0.00514
Margin	Auxiliary contrast	1.256	[1.090, 1.400]	6.34	0.00562
Accuracy	Raw	1.078	[0.993, 1.157]	8.60	0.00514
Accuracy	Gain adjusted	1.097	[0.980, 1.198]	8.31	0.00503
Accuracy	Auxiliary contrast	1.084	[0.939, 1.207]	8.51	0.00516

Per-benchmark reliability averages 0.652 on margins and 0.324 on accuracy. Seven accuracy benchmarks and three margin benchmarks have reliability below 0.50. WinoGrande has a negative estimated margin variance, and both PIQA and WinoGrande have negative accuracy estimates. The accuracy correlation estimate also contains an entry of 1.296 and an eigenvalue of -0.55 , which makes it an invalid correlation matrix without further regularisation. The primary ratio uses covariance products directly, and doesn’t require this normalised matrix to be positive semidefinite.

Removing BoolQ raises margin inflation to 1.786 with interval $[1.689, 1.877]$, and accuracy inflation to 1.558 with interval $[1.412, 1.687]$. Removing CommonsenseQA instead gives inflation estimates of 1.096 and 1.014. BoolQ contributes 68% of the margin covariance trace and 84% of the accuracy trace. The removal estimates imply that the off-diagonal terms involving BoolQ are negative, and they sum to about -0.15 of the margin trace and -0.06 of the accuracy trace. Its removal therefore lowers both the numerator and the denominator of Equation 5, and the denominator falls by more. We recovered each BoolQ question’s passage from the release’s own request files, where the passages of all 3,270 questions match the google/boolq validation split exactly, giving 2,938 passages with 578 questions in shared passages. Assigning whole passages to halves gives inflation of 1.248 and 1.097 on one split. Across fifty re-splits, the passage-aware and item-level rules differ by -0.0002 and -0.0001 on average, with standard deviations 0.0017 and 0.0060. The

BoolQ diagonal moves by 0.03% and 0.15%, so its trace share reflects a large seed variance rather than leaked passage effects. A permutation that reshuffles each run’s item deviations inside every group leaves all trait scores fixed and breaks the tie between particular items and particular runs. Across 2,000 such permutations, 0.087 reach or exceed the observed margin estimate, so item-level sharing accounts for about 0.001 of it.

Recipe removal moves margin inflation between 1.207 and 1.256, and size removal moves it between 1.178 and 1.289, where deleting 530M gives [0.993, 1.338] and stops excluding one. The corresponding accuracy ranges are [1.065, 1.099] and [1.047, 1.115]. Benchmark-removal sensitivity is larger, and comparing that spread with a sampling standard error doesn’t make it a distribution over benchmark collections, since our design treats this battery as fixed. Among equal-weight subsets of the same size, those containing BoolQ average between 0.115 and 0.596 lower margin inflation than those without it, Sweeping 2,000 weightings from a flat Dirichlet leaves 0.817 of margin intervals excluding one, and Appendix C separates benchmark count, identity, and weights.

The defined-trait standardisation diagnostic gives inflation of 1.803 on nine margin traits and 1.707 on eight accuracy traits, with intervals of [1.544, 2.023] and [1.228, 2.076] that treat the estimated weights as fixed. The accuracy value maps to 2.75 effective benchmarks, compared with the full-battery accuracy count of 8.60. This contrast describes a different aggregate, given that both the weights and the retained trait set change. Because the denominators are unstable and we selected the traits on positive estimated variance, we don’t interpret this diagnostic further. Weighting all ten benchmarks by item count instead gives 1.119 with interval [1.068, 1.170] and 1.043 with interval [0.963, 1.119], with MMLU and HellaSwag carrying 64% of the weight.

5.2 FORMAT STRUCTURE AND MECHANISM CHECKS

Among the nine defined margin traits, mean estimated correlations are 0.624 within scoring formats and 0.007 across formats, and Figure 1 shows the full correlation estimates with their eigenvalue spectra. Each benchmark appears in exactly one scoring format, and with that design we can’t separate format from task content causally. Retaining only cross-format off-diagonal covariance gives a diagnostic inflation of 0.921 with interval [0.801, 1.035], compared with 1.244 for the full matrix. This diagnostic isn’t the variance of a separately observed aggregate, because masking covariance entries needn’t preserve positive semidefiniteness. The accuracy diagnostic is 0.965 with interval [0.848, 1.063]. Removing BoolQ gives a cross-format margin diagnostic of 1.006, compared with 1.786 without the restriction.

We also transformed the saved margins item by item, holding items and runs fixed, to test whether the scoring rule alone can produce the gap between the two scales. Dividing each benchmark’s margins by their pooled standard deviation gives margin inflation of 1.182 with interval [1.094, 1.265], bounding them with a hyperbolic tangent gives 1.141 with interval [1.053, 1.222], and keeping only their signs gives 1.079, as Appendix C reports. These are interventions on the scoring rule, and they don’t show how training produces large margins.

The auxiliary-only point estimates retain the full-battery inflation, but their intervals remain wide. The planned signature was margin inflation collapsing after gain adjustment. It didn’t appear, although the proxy limitations restrict any interpretation of that failure. Adjusting for gain increases margin inflation to 1.275 and accuracy inflation to 1.097, joint adjustment gives 1.276 and 1.094, and competence adjustment alone gives 1.244 and 1.076.

The default gain simulation at gain standard deviation 0.05 produces margin inflation of 1.148, about 61% of the observed excess above one. Because the default and distribution-matched simulators produce about 44% and 6% of that excess at the estimated gain standard deviation of 0.042, the share attributed to gain depends on the simulator chosen, and the recipe bootstrap interval for that standard deviation, 0.011 to 0.063, moves the default share from 2% to 91%. The simulations therefore provide no identified causal fraction for sharpness, and we obtain the simulated accuracy invariance from the definition of accuracy alone. The calibrated batch-offset simulation gives factors of 1.004 and 1.003, about 1.5% of the observed margin excess.

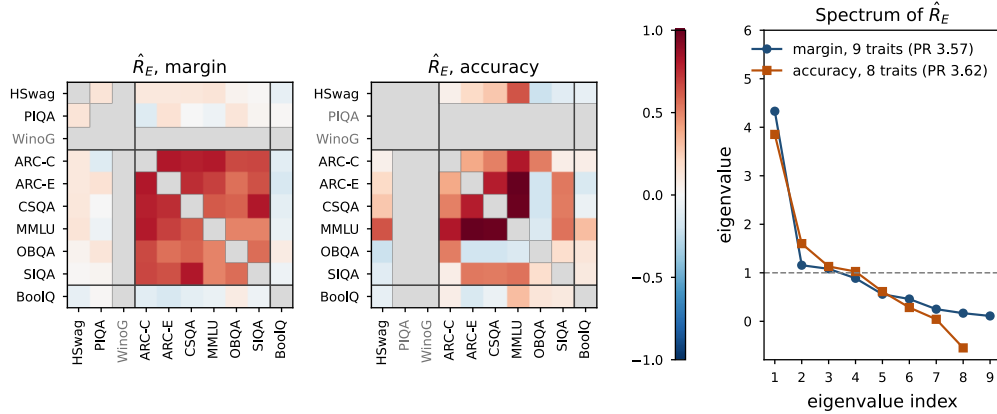


Figure 1. Cross-half correlation estimates across the 125 configurations, with undefined variances shown in grey. The accuracy estimate has a negative eigenvalue, so its spectrum is a diagnostic of the moment estimate rather than a valid population correlation spectrum.

5.3 CALIBRATION AND EXTERNAL CHECKS

Under independent noise, the reported simulation gives mean inflation of 1.0009 on margins and 1.0004 on accuracy. Independent per-trait run-label permutation gives 1.0010 and 0.9984. Across simulated effective covariance coefficients from zero to 0.50, the reported recovery slope is 0.998 with intercept 0.003. Fifty item re-splits have standard deviations of 0.002 and 0.008. These checks assess selected reference distributions, and can’t rule out training-schedule confounding.

The 500-permutation accuracy tail fraction is 0.028, and a later 1,000-permutation run gives 0.021. Both differ from the primary cluster interval, which contains one. We retain both results because permutation exchangeability is stronger than a zero aggregate covariance condition, and because the permutation and cluster procedures condition on different aspects of the design.

PolyPythias gives margin inflation of 1.406 on the three planned sizes, compared with 1.244 for DataDecide. The ratio of excesses above one is 1.66, but the ratio of inflation factors is about 1.13. The reported transfer gate used the first scale after the plan left the comparison ambiguous, so it can’t show failure of a 20% aggregate-standard-deviation criterion. A five-size rescore gives margin inflation of 1.262, with interval $[0.927, 1.537]$, and its same-job three-size estimate is 1.414 with interval $[0.939, 1.781]$. Its three- and five-size accuracy factors are 2.286 and 1.711, and the five-size accuracy interval of $[0.869, 2.428]$ includes DataDecide’s 1.078, while the three-size interval has no defined lower endpoint, and a test-inversion interval with each configuration as a cluster leaves accuracy unbounded at both sizes. Our repeat of this rescore reproduces all four estimates, and the accuracy factors give us a further reason to withhold a general independence recommendation.

The initialisation panel from Heineman et al. (2025) gives likelihood inflation of 1.705 and accuracy inflation of 0.989. Its data-order panel gives corresponding estimates of 1.625 and 0.868. These are single-configuration estimates on different likelihood and accuracy batteries, without cross-half correction or uncertainty intervals. They suggest a score-scale difference, but don’t independently confirm the DataDecide estimand.

5.4 PREDICTING AGGREGATE UNCERTAINTY

We compare five covariance models and one exploratory rescaled plug-in on five held-out recipe folds, giving each model the held-out marginal seed standard deviations, so Table 2 isolates the correlation component, conditional on variance estimates that a practitioner would still need replicate runs to obtain. Before fitting any model, the evaluation clips negative variance estimates, and it retains the original covariance contributions in the target.

The rank-one model has the lowest margin point error, and full covariance has the lowest accuracy point error, but paired recipe-bootstrap intervals with 2,000 refitted draws separate neither model

Table 2. Mean squared prediction error of log aggregate standard deviation across five recipe folds. Appendix C gives paired recipe-bootstrap intervals for differences between these means. The rescaled plug-in is an exploratory addition to the original five-model comparison.

Model	Margin	Accuracy
Independence	0.0358	0.0143
Rank-one competence	0.0057	0.0140
Rank-one plus gain	0.0063	0.0260
Phenotypic correlation plug-in	0.1073	0.1682
Full covariance	0.0068	0.0105
Rescaled phenotypic plug-in	0.0131	0.0342

from its runner-up. On margins, the rank-one, rank-one plus gain, and full models each lower the error relative to independence by about 0.03, and their intervals exclude zero, although the rank-one interval reaches zero when nonpositive held-out variances are filled from training folds, whereas on accuracy every computed interval for a difference from independence includes zero, as Appendix C reports. The unrescaled phenotypic plug-in has the largest error on both scales, and rescaling its off-diagonals on training folds reduces that error, but requires replicate information and still has higher point error than the best model on either scale. An operational comparison on the same folds instead predicts each held-out configuration’s raw aggregate variance moment from covariance matrices refitted without held-out marginals. There, independence has mean squared error 5.01×10^{-9} on margins and 1.72×10^{-9} on accuracy, and the full, rank-one, and shrinkage models differ from it by amounts whose 2,000-draw recipe-resampling intervals all include zero. That target is the noisy three-run moment, so the comparison bounds detectable gains without calibrating the ranking above.

Under independent Gaussian run errors with a common aggregate standard deviation, a true score gap Δ reverses sign with probability

$$P_{\text{flip}}(\Delta) = \Phi\left(-\frac{\Delta}{\sqrt{2}\sigma_{\text{agg}}}\right). \quad (9)$$

Using the reported accuracy deviation of 0.00514 gives probabilities of about 0.246 for a half-point gap and 0.085 for a one-point gap. Under independence, the same gaps give about 0.229 and 0.069. We computed these probabilities from the model without observing reversal rates, and unequal variances require a different denominator. The reported census places 455 of 1,500 observed same-size gaps below the corresponding 5% risk threshold, though noisy gaps give no lower bound on the fraction of small true gaps. The same 1,500 pairs have seed-matched runs, and the paired difference carries a positive variance estimate for 1,445 margin and 1,222 accuracy pairs. For those pairs, the ratio of the paired-difference standard deviation to the independence value has median 1.024 on margins and 1.081 on accuracy, with 5th to 95th percentiles of 0.39 to 1.40 and 0.36 to 1.62, and the stricter subset whose two configurations both carry positive variance holds 1,379 and 919 pairs at medians of 1.027 and 1.079. The seed-matched covariance is positive in 0.49 of pairs, so matching seeds across configurations doesn’t reduce comparison noise in this population.

6 RELATED WORK

Bouthillier et al. (2021) derive variance inflation from correlated repeated trainings of one pipeline on one task and measure the variance from each source in five case studies. We apply covariance accounting across benchmarks within a replicate run. Reimers and Gurevych (2017) and Madaan et al. (2024) study seed-related evaluation variability, and Dodge et al. (2020) examine whether favourable initialisations transfer across tasks. Jordan (2024) separates example-specific training variance from distribution-level variation and compares accuracies across related evaluation sets. Among the studies we located, none estimates the covariance of run noise between benchmarks or an effective benchmark count for a fixed battery. Our contribution is a measurement of that covariance, made with an established split-half construction applied to replicate runs.

The cross-half construction follows repeated-measurement and variance-component ideas from Spearman (1904), Cronbach (1951), and Searle et al. (1992), and the quantitative-genetics analysis

of covariance structure (Martin and Eaves, 1977) provides an analogy, although our model doesn't identify separate genetic or shared-environment components. Cheverud (1988) motivates the phenotypic plug-in comparison in Table 2. Its empirical failure here concerns the transfer of correlation structure between two sources of variation.

Heineman et al. (2025) study evaluation signal and noise, while Magnusson et al. (2025) and van der Wal et al. (2025) provide the replicate populations used here. Dehghani et al. (2021) show the consequences of benchmark composition, which our variance-weighted estimates also expose. Few-cluster inference follows the concerns studied by Cameron et al. (2008) and MacKinnon and Webb (2017), with calibration remaining specific to the implemented procedure and reference population. Appendix B discusses further related approaches and their different units of replication.

7 DISCUSSION AND LIMITATIONS

SNAP estimates cross-half covariance under a measurement model that separates item-general run effects from item-dependent noise. Accuracy inflation is small for the full DataDecide battery, but removing BoolQ raises the estimate, and a true ratio of 1.10 would have been declared in only 0.135 of replicates, so that interval bounds a large ratio rather than showing independence. On margins, the estimate depends on scoring format and gain simulation assumptions, and after these sensitivities we lack sufficient evidence for a general approximation of independent benchmark noise.

Since we used the full dataset without the planned screening split, the empirical study remains exploratory throughout. The passage-aware split bounds BoolQ passage leakage within re-split variation, but common-step scoring leaves differences between training schedules. We floored variances, which let WinoGrande dominate the margin adjustment, and repaired margin proxies built from the same seed deviations fall below one even in simulations without a separate competence variable. The shipped proxy's two item halves correlate at -0.086 on margins, so it carries no measurable signal about a run. We make no claim about the competence mechanism, which needs held-out loss scored on the same checkpoints, and 80 of the 125 model repositories answer an unauthenticated request with HTTP 401, while 40 resolve without one. Simulated coverage of the cluster interval stays between 0.929 and 0.953 across twelve populations, and paired intervals don't separate the best correlation models on either scale. We cluster on recipes, and the release shares one seed label across every recipe inside a size band, so a run effect that follows the size band would break that choice, taking simulated coverage of the recipe-clustered interval to 0.821 when it holds a quarter of latent variance and to 0.580 at a half. A permutation test aligning runs by batch slot bounds that effect at 0.044 of the seed covariance on margins, where simulated coverage is 0.943 rather than the 0.821 a quarter costs. Clustering on the five size bands instead leaves both reported conclusions standing on the observed data, at $[1.030, 1.426]$ for margins and $[0.961, 1.184]$ for accuracy, so the conclusions survive the choice even though the reported widths don't. On PolyPythias, where nine seeds per configuration allow a direct measurement, the decomposition of a three-run fit beats a constant prediction of one when the target pools configurations, and loses to it when the target is one configuration's six-run ratio. Within a single size the evidence is weaker, because the 1B margin estimate of 1.149 has a log-scale interval of 0.811 to 1.628 that includes one, as Appendix C reports.

The PolyPythias experiment, which our repeat rescore reproduces, shows another source of variation in the scored checkpoint. Across nine 160M runs, moving 1,000 steps changes aggregate scores by 0.89 between-seed standard deviations on margins and 1.57 on accuracy. Rescoring 369 DataDecide runs at the previous step shared within each configuration, usually 1,250 steps earlier, shifts them by 0.47 and 0.63 between-seed standard deviations, rising to 0.89 and 1.05 at 750M. On those 123 configurations, margin inflation falls from 1.233 to 1.192 at the earlier step and accuracy inflation rises from 1.082 to 1.130, and the accuracy interval of 1.044 to 1.210 excludes one. Neither change is distinguishable from zero under a paired recipe bootstrap, but the near-independence reading of accuracy depends on the scored checkpoint, as Appendix C reports.

For evaluation practice, we recommend paired replicate scores across the chosen battery, an aggregate seed standard deviation, and a benchmark-removal sensitivity check. A rule assuming benchmark independence declares a margin difference in 0.113 of simulated comparisons with a true gap of zero, and scaling its standard error by a ratio estimated on the same population returns that to 0.051 at any run budget, although carrying that ratio to a battery whose own ratio is 1.5 leaves the error at 0.101, so it has to be measured locally.

AI USE STATEMENT

We used generative AI tools to propose and refine hypotheses, critique the estimator and simulation design, assist with analysis code, revise manuscript text, and survey literature. AI assistance also contributed to the technical and editorial review of this version. The study includes synthetic calibration and sensitivity simulations, whose role differs from the released model outputs used for the empirical analysis. The authors remain responsible for verifying AI-assisted claims, code, citations, and the final submission, including an accurate disclosure of AI involvement in generating simulation data.

ETHICS STATEMENT

This study analyses released model outputs and introduces no new participant recruitment or data collection. Its intended use is to improve uncertainty reporting in model evaluation. An independence approximation used outside the studied battery could understate uncertainty, so the recommendations remain conditional on the measured population and score definition. Use of derived artifacts must retain the applicable source attribution and licensing requirements.

REPRODUCIBILITY STATEMENT

Section 2 and Appendix A define the estimator and its assumptions, while Appendix D specifies the interval calculation and data reduction. Appendix B records the populations and planned checks, and Appendix C reports later sensitivity analyses alongside the primary results. The primary calculation uses 4,999 bootstrap draws, 2,000 calibration replicates, 500 permutation replicates, 50 item re-splits, and master seed 20260101. Reproduction requires the per-item reductions, item ordering and half assignments, selected checkpoint identifiers, and the corresponding analysis scripts. The original analysis plan lacks an independently corroborated timestamp, and its proposed screening split wasn't implemented. The additional simulation, transport, and checkpoint analyses require their own execution records, rather than the primary reduction alone. A separate implementation run on Kaggle reproduces the primary estimates, the BoolQ-removal estimates, and their intervals to three decimals from the released per-item scores, and it produced the passage-aware split, the coverage simulation, the operational prediction comparison, and the seed-matched pair analysis. The PolyPythias transfer, five-size rescore, and checkpoint results come from earlier execution records that this revision didn't rerun, and the loss-proxy rescoring wasn't run.

REFERENCES

- Rishabh Agarwal, Max Schwarzer, Pablo Samuel Castro, Aaron Courville, and Marc Bellemare. Deep reinforcement learning at the edge of the statistical precipice. In *Advances in Neural Information Processing Systems*, volume 34, pp. 29304–29320, 2021.
- Robert M. Bell and Daniel F. McCaffrey. Bias reduction in standard errors for linear regression with multi-stage samples. *Survey Methodology*, 28(2):169–181, 2002.
- Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O'Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. Pythia: A suite for analyzing large language models across training and scaling. In *International Conference on Machine Learning*, volume 202, pp. 2397–2430, 2023.
- Xavier Bouthillier, Pierre Delaunay, Mirko Bronzi, Assya Trofimov, Brennan Nichyporuk, Justin Szeto, Naz Sepah, Edward Raff, Kanika Madan, Vikram Voleti, Samira Ebrahimi Kahou, Vincent Michalski, Dmitriy Serdyuk, Tal Arbel, Chris Pal, Gaël Varoquaux, and Pascal Vincent. Accounting for variance in machine learning benchmarks. In *Proceedings of Machine Learning and Systems (MLSys)*, volume 3, 2021.
- Robert L. Brennan. *Generalizability Theory*. Statistics for Social Science and Public Policy. Springer, New York, 2001.

- William Brown. Some experimental results in the correlation of mental abilities. *British Journal of Psychology*, 3(3):296–322, 1910.
- Nghia Bui, Guergana Savova, and Lijing Wang. Assessing the macro and micro effects of random seeds on fine-tuning large language models. In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (Volume 2: Short Papers)*, pp. 41–46, 2025. arXiv:2503.07329.
- Ryan Burnell, Wout Schellaert, John Burden, Tomer D. Ullman, Fernando Martinez-Plumed, Joshua B. Tenenbaum, Danaja Rutar, Lucy G. Cheke, Jascha Sohl-Dickstein, Melanie Mitchell, Douwe Kiela, Murray Shanahan, Ellen M. Voorhees, Anthony G. Cohn, Joel Z. Leibo, and Jose Hernandez-Orallo. Rethink reporting of evaluation results in AI. *Science*, 380(6641):136–138, 2023.
- A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3):414–427, 2008.
- Donald T. Campbell and Donald W. Fiske. Convergent and discriminant validation by the multitrait-multimethod matrix. *Psychological Bulletin*, 56(2):81–105, 1959.
- Dallas Card, Peter Henderson, Urvashi Khandelwal, Robin Jia, Kyle Mahowald, and Dan Jurafsky. With little power comes great responsibility. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 9263–9274, 2020.
- James M. Cheverud. A comparison of genetic and phenotypic correlations. *Evolution*, 42(5):958–968, 1988.
- James M. Cheverud. A simple correction for multiple comparisons in interval mapping genome scans. *Heredity*, 87(1):52–58, 2001.
- Lee J. Cronbach. Coefficient alpha and the internal structure of tests. *Psychometrika*, 16(3):297–334, 1951.
- Mostafa Dehghani, Yi Tay, Alexey A. Gritsenko, Zhe Zhao, Neil Houlsby, Fernando Diaz, Donald Metzler, and Oriol Vinyals. The benchmark lottery. arXiv:2107.07002, 2021.
- Janez Demšar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7:1–30, 2006.
- Jesse Dodge, Gabriel Ilharco, Roy Schwartz, Ali Farhadi, Hannaneh Hajishirzi, and Noah Smith. Fine-tuning pretrained language models: Weight initializations, data orders, and early stopping, 2020. arXiv:2002.06305.
- Rotem Dror, Gili Baumer, Marina Bogomolov, and Roi Reichart. Replicability analysis for natural language processing: Testing significance with multiple datasets. *Transactions of the Association for Computational Linguistics*, 5:471–486, 2017.
- Nicolas Dufour, Alexei A. Efros, and Patrick Pérez. The FID lottery: Quantifying hidden randomness in generative-model evaluation. arXiv:2606.20536, 2026.
- Kyle Gorman and Steven Bedrick. We need to talk about standard splits. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pp. 2786–2791, 2019.
- Michael Hagmann, Philipp Meier, and Stefan Riezler. Towards inferential reproducibility of machine learning research. In *International Conference on Learning Representations*, 2023.
- Michael Hardy, Anka Reuel, Lijin Zhang, Jodi M. Casabianca, Sang Truong, Yash Dave, Hansol Lee, Benjamin Domingue, and Sanmi Koyejo. AI cartography: Mapping the latent landscape of AI benchmark ecosystems. arXiv:2605.25272, 2026.
- Andreas Haupt, Justin Hartenstein, Anka Reuel, Mykel Kochenderfer, and Sanmi Koyejo. Welfare, improvability, and variance: A principal-agent approach to optimal benchmark item aggregation. arXiv:2605.30916, 2026.

- Larry V. Hedges, Elizabeth Tipton, and Matthew C. Johnson. Robust variance estimation in meta-regression with dependent effect size estimates. *Research Synthesis Methods*, 1(1):39–65, 2010.
- David Heineman, Valentin Hofmann, Ian Magnusson, Yuling Gu, Noah A. Smith, Hannaneh Hajishirzi, Kyle Lo, and Jesse Dodge. Signal and noise: A framework for reducing uncertainty in language model evaluation. In *Advances in Neural Information Processing Systems*, 2025. arXiv:2508.13144.
- Peter Henderson, Riashat Islam, Philip Bachman, Joelle Pineau, Doina Precup, and David Meger. Deep reinforcement learning that matters. *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1):3207–3214, 2018.
- Andreas Hochlehnert, Hardik Bhatnagar, Vishaal Udandaraao, Samuel Albanie, Ameya Prabhu, and Matthias Bethge. A sober look at progress in language model reasoning: Pitfalls and paths to reproducibility. In *Conference on Language Modeling (COLM)*, 2025.
- Valentin Hofmann, David Heineman, Ian Magnusson, Kyle Lo, Jesse Dodge, Maarten Sap, Pang Wei Koh, Chun Wang, Hannaneh Hajishirzi, and Noah A. Smith. Fluid language model benchmarking, 2025. arXiv:2509.11106.
- Keller Jordan. On the variance of neural network training with respect to test sets and distributions. In *International Conference on Learning Representations*, 2024.
- Siddharth Karamcheti, Laurel Orr, Jason Bolton, Tianyi Zhang, Karan Goel, Avani Narayan, Rishi Bommasani, Deepak Narayanan, Tatsunori Hashimoto, Dan Jurafsky, Christopher D. Manning, Christopher Potts, Christopher Ré, and Percy Liang. Mistral: A Journey towards Reproducible Language Model Training. Stanford Center for Research on Foundation Models, 2021. <https://crfm.stanford.edu/2021/08/26/mistral.html>.
- Alex Kipnis, Konstantinos Voudouris, Luca M. Schulze Buschoff, and Eric Schulz. metabench, a sparse benchmark of reasoning and knowledge in large language models, 2024. arXiv:2407.12844.
- Leslie Kish. *Survey Sampling*. John Wiley & Sons, New York, 1965.
- Guneet Kohli. Nine judges, two effective votes: Correlated errors undermine LLM evaluation panels. arXiv:2605.29800, 2026.
- J. Li and L. Ji. Adjusting multiple testing in multilocus analyses using the eigenvalues of a correlation matrix. *Heredity*, 95(3):221–227, 2005.
- James G. MacKinnon and Matthew D. Webb. Wild bootstrap inference for wildly different cluster sizes. *Journal of Applied Econometrics*, 32(2):233–254, 2017.
- Lovish Madaan, Aaditya K. Singh, Rylan Schaeffer, Andrew Poulton, Sanmi Koyejo, Pontus Stenertorp, Sharan Narang, and Dieuwke Hupkes. Quantifying variance in evaluation benchmarks. arXiv:2406.10229, 2024.
- Ian Magnusson, Nguyen Tai, Ben Bogin, David Heineman, Jena D. Hwang, Luca Soldaini, Akshita Bhagia, Jiacheng Liu, Dirk Groeneveld, Oyvind Tafjord, Noah A. Smith, Pang Wei Koh, and Jesse Dodge. DataDecide: How to predict best pretraining data with small experiments. In *International Conference on Machine Learning*, 2025. arXiv:2504.11393.
- N. G. Martin and L. J. Eaves. The genetical analysis of covariance structure. *Heredity*, 38(1):79–95, 1977.
- R. Thomas McCoy, Junghyun Min, and Tal Linzen. BERTs of a feather do not generalize together: Large variability in generalization across models with similar test set performance. In *Proceedings of the Third BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP*, pp. 217–227, 2020.
- Solomon Messing. Hidden measurement error in LLM pipelines distorts annotation, evaluation, and benchmarking. arXiv:2604.11581, 2026.

- Evan Miller. Adding error bars to evals: A statistical approach to language model evaluations. arXiv:2411.00640, 2024.
- Dale R. Nyholt. A simple correction for multiple testing for single-nucleotide polymorphisms in linkage disequilibrium with each other. *American Journal of Human Genetics*, 74(4):765–769, 2004.
- Nils Reimers and Iryna Gurevych. Reporting score distributions makes a difference: Performance study of LSTM-networks for sequence tagging. In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, pp. 338–348, 2017.
- Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. Are emergent abilities of large language models a mirage? In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- Shayle R. Searle, George Casella, and Charles E. McCulloch. *Variance Components*. John Wiley & Sons, New York, 1992.
- Thibault Sellam, Steve Yadlowsky, Ian Tenney, Jason Wei, Naomi Saphra, Alexander D’Amour, Tal Linzen, Jasmijn Bastings, Iulia Turc, Jacob Eisenstein, Dipanjan Das, and Ellie Pavlick. The MultiBERTs: BERT reproductions for robustness analysis. In *International Conference on Learning Representations*, 2022.
- Tommy Sha and Stella Zhao. BenchScope: How many independent signals does your benchmark provide? arXiv:2603.29357, 2026.
- C. Spearman. The proof and measurement of association between two things. *The American Journal of Psychology*, 15(1):72–101, 1904.
- C. Spearman. Correlation calculated from faulty data. *British Journal of Psychology*, 3(3):271–295, 1910.
- Team OLMo, Pete Walsh, Luca Soldaini, Dirk Groeneveld, Kyle Lo, et al. 2 OLMo 2 furious. arXiv:2501.00656, 2025. Shorter version in *Conference on Language Modeling (COLM)* 2025.
- Oskar van der Wal, Pietro Lesci, Max Müller-Eberstein, Naomi Saphra, Hailey Schoelkopf, Willem Zuidema, and Stella Biderman. PolyPythias: Stability and outliers across fifty language model pre-training runs. In *International Conference on Learning Representations*, 2025. arXiv:2503.09543.
- Yang Zhao, Jiahao Lu, Bin Huang, Guhua Zhang, and Jie Zhou. Most transformer modifications still do not transfer at 1–3b: A 2020–2026 update to Narang et al. (2021) with downstream evaluation and a noise floor. arXiv:2605.20798, 2026.
- Xiang Zhou, Yixin Nie, Hao Tan, and Mohit Bansal. The curse of performance instability in analysis datasets: consequences, source, and suggestions. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 8215–8228, 2020.

A DERIVATIONS AND INTERPRETATION

A.1 WHAT SNAP ESTIMATES

SNAP abbreviates Seed Noise Across Phenotypes, with a phenotype denoting a benchmark score. Its target is the covariance of run deviations that generalise across item halves. DataDecide replicates vary initialisation and data order together, and auxiliary runs below 1B use shortened training schedules. Common-step scoring aligns the observation step, while schedule differences remain within the full-set target. We therefore use run covariance for the empirical estimate, reserving a seed-only interpretation for designs with matching training conditions.

The item-general estimate can differ from the run variance of a fixed evaluation bank. A run can perform differently on particular items without changing its expected score over an item population. Under the measurement model, that variation enters the item error and doesn't contribute to the cross-half covariance target. Repeated evaluation of a fixed item bank can retain this component, so its run variance needn't equal the item-general variance estimated here.

The analogy with quantitative genetics distinguishes covariance between configurations from residual covariance within them. It doesn't identify separate additive-genetic and shared-environment components in a model-training experiment. Between-configuration correlation R_P reflects differences in recipe and model size, while the within-configuration covariance Σ_E concerns replicate deviations. The plug-in comparison asks whether the first correlation structure predicts the second after supplying marginal run variances.

A.2 NOTATION

Table 3. Notation for the cross-half construction and its empirical diagnostics. The coefficient $\bar{r}_E$ uses covariance normalisation, which differs from mean correlation under unequal marginal variances.

Symbol	Meaning
c, r, j, i	Configuration, replicate run, benchmark, and item indices
N, R, K	Configuration, replicate, and benchmark counts, with values 125, 3, and 10
$\Sigma_{E,c}, \Sigma_E$	Configuration covariance and its average across the design
R_E, R_P	Run-effect correlation and between-configuration score correlation
y^M, y^A	Mean per-byte margin and accuracy on the specified item set
$d_{cr,j}^H, g_{cr}^H$	Centred benchmark score and its benchmark average on half H
T_c, U_c	Cross-half aggregate covariance and its diagonal-only contribution
$\Lambda, \bar{r}_E$	Standard-deviation inflation and effective covariance coefficient
K_{eff}	Independent-benchmark equivalent at the same marginal RMS deviation
$\sigma_{\text{agg}}, \sigma_{\text{indep}}$	Aggregate deviation with full covariance and under independence
$\rho_g, \bar{v}$	Aggregate reliability and its scaled item-noise contribution
$s_{cr}, x_{cr}^{(-j)}$	Gain covariate and opposite-half competence proxy

A.3 CROSS-HALF IDENTITY AND ITS ASSUMPTIONS

Fix a configuration and suppress its index for clarity. Write the half-score model as $y_{rj}^H = \mu_j^H + E_{rj} + \varepsilon_{rj}^H$, so centring across runs gives

$$d_{rj}^H = (E_{rj} - \bar{E}_{\cdot j}) + (\varepsilon_{rj}^H - \bar{\varepsilon}_{\cdot j}^H). \quad (10)$$

Assume that run-effect vectors are independent draws with covariance Σ_E . Conditional on those effects, assume zero-mean item errors and zero covariance between ε_{rj}^A and ε_{sk}^B for the run and benchmark pairs in the calculation. Expanding $d_{rj}^A d_{rk}^B$ then removes its error cross terms in expectation, leaving

$$\mathbb{E}[d_{rj}^A d_{rk}^B] = \mathbb{E}[(E_{rj} - \bar{E}_{\cdot j})(E_{rk} - \bar{E}_{\cdot k})] \quad (11)$$

$$= (1 - 1/R)\Sigma_E(j, k). \quad (12)$$

Summing across runs and dividing by $R - 1$ gives the unbiased covariance moment. We symmetrise the two half orderings for the matrix estimate,

$$\hat{\Sigma}_E(j, k) = \frac{1}{2N(R - 1)} \sum_{c, r} (d_{crj}^A d_{crk}^B + d_{crk}^A d_{crj}^B). \quad (13)$$

Symmetry follows from this construction, while positive semidefiniteness doesn't follow for a finite sample. A negative diagonal moment near zero therefore doesn't by itself falsify the identity, although it prevents ordinary correlation normalisation for that trait.

Different item identifiers can still refer to items with dependent errors. Shared source passages, duplicate questions, and overlapping content can create cross-half covariance. A fixed finite item bank split without replacement also requires a model for its sampling dependence. The available item-half assignment alone can't verify these conditions.

BoolQ has documented shared passages, so we check it directly. The release's request files carry each question's passage, and the passages of all 3,270 questions match the google/boolq validation split exactly, giving 2,938 passages with 578 questions in passages of two or more. A passage-aware split assigns whole passages to halves and then averages questions, so half sizes differ by a few items and the equal-half derivation doesn't hold exactly. On one such split the margin and accuracy estimates are 1.248 and 1.097 against 1.244 and 1.078 on the frozen split, and across fifty paired re-splits the passage-aware and item-level rules differ by -0.0002 and -0.0001 on average with standard deviations 0.0017 and 0.0060. The BoolQ diagonal moves by 0.03 percent and 0.15 percent. A simulation matched to the observed passage-size histogram, with 500 replicates at each of five interaction scales, adds a run-by-passage interaction shared by the questions on a passage, with standard deviation from zero up to the item-noise scale. Across those scales the item-level and passage-level splits differ in Λ by between -0.0008 and 0.0022 on average, within Monte Carlo error of zero, because the observed histogram puts about 240 same-passage question pairs across the halves out of about 2.7 million cross-half pairs, so a leaked interaction can't move the diagonal by more than about 0.01 percent even at that scale. The null result above is therefore what the passage structure of this benchmark implies rather than evidence of a sensitive test. Outside BoolQ, the request files show two question stems shared across traits and repeated question stems within benchmarks, for example 62 of 1,838 PIQA items, which we haven't grouped.

For a diagonal additional covariance D with nonnegative trace, the inflation ratio becomes

$$\Lambda_D^2 = \frac{\mathbf{1}^T \Sigma_E \mathbf{1} + \text{tr } D}{\text{tr } \Sigma_E + \text{tr } D}. \quad (14)$$

This ratio lies between Λ^2 and one when the denominators are positive. It explains attenuation toward one under that restricted model, but doesn't yield a general upper or lower bound when shared-item effects also contribute off-diagonal covariance. The same restriction is needed before using the item-general factor to bound fixed-battery inflation.

A.4 COVARIANCE COEFFICIENT AND EFFECTIVE COUNT

Let $\sigma_j^2 = \Sigma_E(j, j)$ and $r_{jk} = \Sigma_E(j, k) / (\sigma_j \sigma_k)$ whenever both variances are positive. For this finite battery, the variance ratio therefore satisfies

$$\Lambda^2 = 1 + \frac{\sum_{j \neq k} r_{jk} \sigma_j \sigma_k}{\sum_j \sigma_j^2} = 1 + (K - 1) \bar{r}_E. \quad (15)$$

The normalised weighted mean correlation instead divides the numerator by $\sum_{j \neq k} \sigma_j \sigma_k$. For a positive denominator, the two coefficients are related by

$$\bar{r}_E = a_\sigma \bar{r}_{\text{weighted}}, \quad a_\sigma = \frac{\sum_{j \neq k} \sigma_j \sigma_k}{(K - 1) \sum_j \sigma_j^2}. \quad (16)$$

For nonnegative standard deviations, a_σ lies between zero and one. With two perfectly correlated traits having standard deviations one and two, the effective coefficient is 0.8 even though the weighted mean correlation is one. Equal marginal variances make the coefficient and the mean correlation coincide.

With average marginal variance $\bar{\sigma}_E^2 = K^{-1} \text{tr } \Sigma_E$, the variance of the benchmark average satisfies

$$\sigma_{\text{agg}}^2 = \bar{\sigma}_E^2 \left[\frac{1}{K} + \frac{K-1}{K} \bar{r}_E \right]. \quad (17)$$

A nonzero asymptotic floor follows only for a sequence of batteries whose average variance and effective coefficient converge to suitable limits. Bounded variances alone don't imply that convergence, and the present ten-benchmark study doesn't estimate such a sequence. We use the finite-battery identity for the reported effective counts.

For the raw, gain-adjusted, and auxiliary-contrast rows, the reported effective margin coefficients are 0.0608, 0.0695, and 0.0641. The corresponding accuracy coefficients are 0.0181, 0.0225, and 0.0194, retaining the covariance normalisation of Equation 6.

For positive aggregate variance, $K_{\text{eff}} = K/\Lambda^2$ compares the measured average with an independent average having the same marginal RMS deviation. Negative aggregate off-diagonal covariance can give an effective count above the benchmark count, $K_{\text{eff}} > K$. A single positive-variance benchmark has $\Lambda = 1$, although its off-diagonal coefficient is undefined because $K - 1 = 0$. This arithmetic count also differs from a spectral participation ratio.

A.5 POOLING AND FINITE-SAMPLE BIAS

We pool T_c and U_c across configurations before taking their ratio. Each individual denominator uses two replicate degrees of freedom and can approach zero or become negative. Averaging individual ratios would give those unstable denominators disproportionate influence. Pooling reduces that problem but doesn't remove the effect of configurations with large covariance contributions.

Recomputing the recipe influences from the cluster linearisation gives participation ratios of 3.08 for margins and 9.02 for accuracy. One recipe contributes 0.520 of the squared margin influence, and removing that recipe moves the estimate by 0.037, which is the largest change any single recipe produces. A leverage-corrected influence, which divides each recipe residual by one minus that recipe's share of the denominator sum, raises the share to 0.538 and lowers the two ratios to 2.92 and 8.56, so this concentration figure depends on the correction a reader assumes. These quantities describe influence concentration, and aren't literal counts of independent clusters for inference. Recomputing the permutation experiment ten times at the record's own 1,000 replicates gives margin coverage between 0.915 and 0.942 with a mean of 0.931, so the recorded 0.925 sits inside the seed-to-seed spread and the shortfall it reported does reproduce.

The covariance moments are unbiased under their assumptions, while the pooled ratio has a random denominator. Its square root introduces another nonlinear transformation, so neither transformation inherits exact unbiasedness. A finite implementation must also specify what happens when a pooled sum is nonpositive. The reported data give finite full-battery estimates, but that observation doesn't establish finite estimates for arbitrary input populations.

A.6 PERMUTATION REFERENCE

For each configuration and trait, apply a uniform permutation π_j of run labels to both halves together. The diagonal products then sum to the same value, leaving U_c unchanged. Independent permutations across traits give zero expected off-diagonal products because each centred deviation sums to zero. Conditional on a positive fixed pooled denominator, this implies

$$\mathbb{E}_\pi \left[\frac{\sum_c T_c^\pi}{\sum_c U_c} \right] = 1. \quad (18)$$

This expectation for the squared ratio doesn't imply that $\mathbb{E}_\pi[\hat{\Lambda}^\pi] = 1$. If all permuted ratios are non-negative, concavity of the square root gives an expectation no larger than one, with strict inequality for a nonconstant ratio. For example, equally likely ratios of 0.5 and 1.5 average to one, while their square roots average to about 0.966.

A common permutation across all traits preserves their pairings and can't remove cross-trait association. Independent trait permutations avoid that problem, and the identity belongs in the uniform permutation sample space. Its chance of reproducing the observation doesn't invalidate a permutation procedure. The construction preserves marginal score values and cluster membership, but can

alter dependence of run alignments across configurations, so validity as a test requires an exchangeability assumption beyond zero mean covariance.

A.7 PLANNING CALCULATIONS AND INFORMATION RATIO

The author-reported power calculation assumes approximately Gaussian half-score deviations. With $d^A = e + a$, $d^B = e + b$, independent errors a, b , and $\text{Var}(a) = \text{Var}(b) = k \text{Var}(e)$, the product variance is

$$\text{Var}(d^A d^B) = \sigma^4[(1+k)^2 + 1], \quad k = 2(1 - \rho_g)/\rho_g. \quad (19)$$

The planning values used two replicate degrees of freedom per configuration, numerator-denominator correlation 0.7, and design effect 1.60. Appendix B preserves the resulting values for the intended 85-configuration analysis, which differs from the realised 125-configuration study. We don't infer realised power from the observed effect.

For the information calculation, let item margins follow a Gaussian location model with standard deviation s_m . An infinitesimal shift δ changes accuracy by $f(0)\delta + o(\delta)$, while it changes the mean margin by δ . Under independent item sampling, the item-noise variances are $p(1-p)/n$ and s_m^2/n , and the Gaussian model gives $f(0) = \phi(z_0)/s_m$. Dividing noise by squared local sensitivity yields Equation 7. Both the scale cancellation and minimum $\pi/2$ require the Gaussian location model used in this calculation.

The competence and gain adjustments have additional measurement limitations. The opposite-half competence proxy uses estimated seed standard deviations, including a 10^{-6} floor where the margin variance estimate for WinoGrande is -3.15×10^{-7} . The gain covariate uses both halves, so its item noise can correlate with the outcome errors. The reported zero-mediation simulation treats this shared item component differently from the observed regression. Its margin reference begins at 1.24502 rather than the observed 1.24395, and moves to 1.0910 with standard deviation 0.096, whereas the fully adjusted observed value is 1.276. The accuracy reference begins at 1.11831 rather than the observed 1.07837, and ends at 1.1045 against an adjusted 1.094. These mismatches prevent a calibrated causal interpretation of the residualisation comparisons.

B DESIGN DETAILS

B.1 COMMON STEP RULE

For each of the 125 configurations, we selected the largest released checkpoint step available for all three runs. The analysis records that step in each reduced filename, although the supplement described in the analysis record omits the fetch stage's complete configuration table. In 92 configurations, the three runs have different maximum steps. DataDecide stops the two auxiliary runs below 1B at 25 percent of the 1B compute budget and trains three complete runs only at 1B (Magnusson et al., 2025). The per-instance release contains auxiliary runs from 150M upward, so the five replicated sizes used here begin at 150M rather than 4M. This availability rule explains the difference from DataDecide's fourteen-size ladder.

In 26 configurations, one at 530M and 25 at 750M, the largest maximum step exceeds the smallest by more than a factor of 1.5. We indexed the member names of the 25 released recipe archives to recover every run's maximum step, and the index reproduces the selected step recorded for all 375 runs. At 530M the selected step averages 86.7 percent of the default run's final step, with configuration values from 38.9 to 88.7 percent. At 750M it averages 41.5 percent, with values from 17.7 to 44.0 percent. Only 33 configurations have three runs that share a final step, and 24 of those sit at 1B.

Table 4 recomputes both estimates on subsets defined by the index. Excluding the 26 severely truncated configurations raises margin inflation to 1.276 and accuracy inflation to 1.120, and neither wild interval includes one. The 33 configurations with a shared final step give intervals too wide to separate inflation from one, and 18 of their 24 recipes contribute only a 1B configuration. We also define a configuration's influence as $(T_c - \hat{\Lambda}^2 U_c) / \sum_{c'} U_{c'}$, which sums to zero and measures how far the configuration pulls the squared ratio. Its Spearman correlation with the selected step's share of the default final step is -0.036 for margins and 0.001 for accuracy. Permuting that share

within each size over 4,999 draws gives $p = 0.314$ and $p = 0.990$. Within the two truncated bands, which hold 25 configurations each, the correlations are -0.011 and 0.155 for margins and 0.275 and 0.232 for accuracy, and we didn't test those band values separately. We chose these subsets after seeing the full-sample estimates, so they describe sensitivity and don't test a stated hypothesis. The auxiliary-run contrast and the estimates by size in Table 9 give two further checks.

Table 4. Inflation estimates on subsets defined by each configuration's checkpoint schedule, with wild cluster bootstrap-t intervals over 4,999 draws. Severe truncation means that the largest final step among the three runs exceeds the smallest by more than a factor of 1.5.

Subset	Configurations	Recipes	Margin		Accuracy	
All	125	25	1.244	[1.143, 1.338]	1.078	[0.993, 1.157]
Shared final step	33	24	1.082	[0.121, 1.530]	1.129	[0.769, 1.397]
Without severe truncation	99	25	1.276	[1.114, 1.420]	1.120	[1.020, 1.208]
Severe truncation only	26	25	1.107	[0.692, 1.397]	0.977	[0.357, 1.306]
Different final steps	92	25	1.330	[1.024, 1.583]	1.070	[0.956, 1.171]

B.2 RELIABILITY PHENOTYPES

The item halves contain 18,840 and 18,842 items across the battery. The fetch stage creates a balanced split within each trait and records the split for reuse across runs. The planning document proposed estimating reliability on eight screening recipes before inspecting seventeen estimation recipes. We didn't implement that separation, and Table 5 instead uses all 25 recipes. The reliability checks and the decisions based on them therefore remain exploratory.

Table 5. Cross-half reliability of run deviations across all 25 recipes. Reliability equals $\hat{\sigma}_E^2 / (\hat{\sigma}_E^2 + \hat{\sigma}_{\text{noise}}^2)$ and can be negative when the estimated run component is negative. MC denotes a multiple-choice scoring format in this table.

Trait	Format	Accuracy	Margin
ARC-Easy	MC	0.602	0.927
ARC-Challenge	MC	0.203	0.584
BoolQ	yes/no	0.970	0.993
CommonsenseQA	MC	0.615	0.930
HellaSwag	cloze	0.450	0.757
MMLU	MC	0.022	0.860
OpenBookQA	MC	0.091	0.392
PIQA	cloze	-0.028	0.273
Social IQa	MC	0.416	0.888
WinoGrande	cloze	-0.102	-0.086
Battery mean		0.324	0.652

Two accuracy traits have negative reliability estimates, and two others fall below 0.10. Recomputing the per-trait information ratios gives a median of 1.608, a minimum of 1.571, and a maximum of 1.732, which reproduces the three figures the pipeline summary reported without releasing the ratios themselves. The battery-level ρ_g recomputes to 0.961 on margins and 0.734 on accuracy, against the planning value of 0.49 at $\Lambda^2 = 1.819$. The reported matrices contain the mean variance used to reconcile those quantities.

B.3 POWER CALCULATIONS

Table 6 reproduces the planning arithmetic for 85 configurations and seventeen recipe clusters. The completed analysis instead uses 125 configurations and 25 clusters. On the Λ scale, the reported linearised standard errors are 0.045 for margins and 0.038 for accuracy, compared with planning values of 0.0935 and 0.1232. The corresponding variance-ratio standard errors, 0.112 and 0.081, belong to the Λ^2 scale and aren't comparable with the planning column.

The realised standard errors are smaller by factors of 2.1 and 3.3. Increasing the configuration count alone would predict a factor of approximately 1.21 under otherwise fixed assumptions. Interpolation of the N1 simulation grid gives dispersion at the planned $\bar{r}_E = 0.0911$ that is 1.04 times the dispersion at the observed margin value of 0.061, and 1.13 times the dispersion at the accuracy value of 0.0181. Differences in the design effect and reliability also affect this comparison, but we haven't separated their contributions. We retain the original planning calculations rather than calculating power from the estimated effect after observing the results.

Table 6. Planning calculations at $\Lambda^2 = 1.819$ and $N = 85$ configurations. The calculations use $t(16)$, a design effect of $1 + 4\rho_{\text{ICC}} = 1.60$ at $\rho_{\text{ICC}} = 0.15$, and $\rho_g = 0.61$ for margins or 0.49 for accuracy. Power entries are reported from the planning document, which doesn't record the alternative hypothesis they were computed against.

Phenotype	Aggregation	Planned SE(Λ)	MDE(Λ)	Power
Margin	Cross-half	0.0935	1.300	0.89
Margin	Battery mean	0.0845	n/a	0.94
Accuracy	Cross-half	0.1232	1.412	0.68
Accuracy	Battery mean	0.1076	n/a	0.79
Accuracy	Cross-half, $\bar{r}_E = 0.115$	0.1232	n/a	0.82

B.4 DECISION CHECKS AND DEVIATIONS

The planning document lists nine checks with proposed responses. Its claimed date is September 5, 2026, but we haven't established that date through an independent timestamp. We consequently describe the document as a reported plan and treat its checks as exploratory assessments.

A batch offset can contribute to the same cross-trait products as run-level variation. The auxiliary-run contrast removes the shared default-versus-auxiliary contrast from that estimate, while the estimate using all three runs remains exposed. Check G8 passes on point estimates, although the intervals cross the decision boundaries. We haven't corrected the full-set estimate for this exposure.

Under the default gain simulator, $\sigma_g = 0.05$ produces approximately 61 percent of the observed margin excess. An empirically matched simulator produces approximately eight percent at the same gain scale and six percent at the estimated scale of 0.042. These results describe sensitivity to simulator assumptions and aren't causal bounds on the contribution of gain. The accuracy result offers no independent protection against this mechanism because multiplying a margin by a positive factor leaves its sign unchanged. Partialling the gain covariate increases Λ_M , and adding gain to the rank-one predictor worsens held-out point error relative to the scalar predictor alone. These observations don't identify a unique causal explanation.

The analysis didn't execute the proposed responses for G3 and G5. G3 combines asymmetry with negative variance estimates, although a negative moment estimate doesn't establish leakage. The available records also lack a usable source-document identifier for the proposed alternative split. We retain the margin analysis after the G5 failure with the simulator sensitivity disclosed, which departs from the proposed response. The source gate table contains seven rows, with transport and the auxiliary-run contrast supplying G6 and G8 separately.

B.5 REPORTED PLANNING PREDICTIONS

The planning document specifies eleven predictions and their rejection conditions. Table 8 records the outcomes without treating the document as verified preregistration.

- Both $\Lambda_M > 1.349$ and $\Lambda_A > 1.40$, with lower confidence limits above one, constitute the first planned prediction.
- The accuracy estimate retains at least 80 percent of the margin estimate, expressed as $\Lambda_A \geq 0.8\Lambda_M$.
- Accuracy retains more than 70 percent of its excess after gain adjustment, and N5 generates less than 30 percent of the margin excess.

Table 7. Planning checks, proposed responses, and observed outcomes in the source analysis. The original assessment labels four checks as failures, while G2 uses a lower threshold that can’t discriminate under its model.

Check	Condition and timing	Proposed response	Reported outcome
G0	Outputs parse at the common step, day 1	Drop unavailable cells, report reduced N if fewer than 70 estimation cells remain	Pass, all 125 cells
G1	Accuracy power at $\bar{r}_E = 0.115$ at least 0.80, day 2	Make accuracy secondary and report margin in the abstract	Pass, power 1.00 at $\Lambda_{\text{alt}} = 1.427$, $\rho_g = 0.73$
G2	Median information ratio at least 1.4, day 2	Remove comparative information claim	Ratio 1.61, threshold below the ratio’s floor
G3	Symmetric cross-half covariance and positive diagonal, day 3	Split by source document, then report leakage if asymmetry persists	Fail, one negative margin diagonal and two accuracy diagonals
G4	$ \hat{\Lambda} - 1 \leq 0.02$ under N1 and N2, day 4	Subtract calibrated offset and report both estimates	Pass, offset at most 0.002
G5	N5 accounts for less than 50 percent of excess, day 5	Lead with sharpness and report accuracy alone	Fail, default simulator share 0.61
G6	PolyPythias transport error within 20 percent, day 6	Report transport failure and its ratio	Fail upward under the excess-ratio reading
G7	Held-out loss available in metadata, day 1	Use leave-one-trait-out competence covariate	Fail, no required field
G8	Auxiliary-run $\Lambda \geq 1 + 0.5(\Lambda_{\text{full}} - 1)$, day 5	Lead with auxiliary-run estimate if condition fails	Pass on both point estimates, margin 1.256 versus 1.122, intervals cross their boundaries

4. At measured token-budget ratios, N6 generates less than 50 percent of the full-set excess.
5. The independence calculation underpredicts aggregate seed standard deviation by Λ in at least four of five size bands.
6. DataDecide’s R_E predicts PolyPythias aggregate seed standard deviation within 20 percent, while independence misses by more than 20 percent.
7. At least one seed mechanism in the external panel gives $\Lambda > 1$ and a bootstrap lower limit above one.
8. At least eight traits have an information ratio between 1.4 and 3.0 in the planned assessment.
9. The cross-format diagnostic exceeds 1.2, with its lower confidence limit also exceeding one.
10. The Frobenius distance $\|R_E - R_P\|_F$ exceeds 0.5 in the comparison between the two correlation matrices.
11. More than ten percent of same-size configuration pairs have $P_{\text{flip}} > 0.05$ at the measured aggregate noise scale.

B.6 EXCLUDED POPULATIONS

MultiBERTs provides 25 seeds for one configuration (Sellam et al., 2022), whereas the 100 BERT runs of McCoy et al. (2020) vary fine-tuning from one pretrained checkpoint. Pythia’s size ladder lacks seed replicates (Biderman et al., 2023), and the OLMo-2 checkpoints considered here lack replicates at matched steps (Team OLMo et al., 2025). The signal-and-noise panel lacks per-item outputs, which limits its use to the external check. We exclude these populations because they lack the matched replicates or item outputs required here.

B.7 RELATION TO PRIOR MEASUREMENT WORK

Reimers and Gurevych (2017) compare score distributions across seeds within datasets, while Madaan et al. (2024) report benchmark-level seed variance. Hagmann et al. (2023) model non-determinism with scalar variance components and their interactions with data properties. Our target adds covariance between benchmark-specific run deviations to that measurement problem.

Jordan (2024) attributes much test-set variance to per-example variation across seeds, and its Appendix C compares run accuracy across five ImageNet evaluation sets with R^2 at most 0.14. Those comparisons concern shifted versions of one benchmark. Bui et al. (2025) vary fine-tuning seeds, and Hochlehnert et al. (2025) vary decoding seeds, which address different variance components from the pretraining-run variation examined here. Dodge et al. (2020, Section 4.3) ask whether a seed performs well across datasets and compare four GLUE task rankings without estimating a covariance matrix.

Zhou et al. (2020, Figure 3) report a Spearman correlation matrix across datasets. Their standard block has correlations at least 0.95, while standard-to-analysis entries include 0.97 for STR-L and 0.30 for STR-O. These correlations compare checkpoints within runs and average across seeds, so they combine changes in checkpoint performance rather than isolating within-configuration seed deviations. Dufour et al. (2026) separate training and sampling variation in FID, while Miller (2024) estimates item-level uncertainty for one model. Zhao et al. (2026) report the three-seed standard deviation of a twelve-task average and combine twelve per-task Welch tests in their Appendix E.4 without estimating cross-task covariance.

Haupt et al. (2026) use the PolyPythias 410M panel to estimate per-item variation and prioritise items by welfare weight and improvability. Their per-item variance estimates and approximate item-independence assumption address a different quantity from the off-diagonal covariance of benchmark run effects. Their item-level estimates and our benchmark covariance therefore answer different questions using some of the same models.

Cheverud (2001), Nyholt (2004), and Li and Ji (2005) estimate effective numbers of independent tests from observed correlation matrices. Cheverud uses eigenvalue variance, Nyholt uses spectral decomposition of linkage disequilibrium, and Li and Ji revise the effective-count approximation. Their application adjusts significance thresholds, whereas ours converts an estimated noise covariance into an aggregate standard deviation and an effective benchmark count. Searle et al. (1992) explain negative variance-component estimates from moment methods when the true component lies near zero.

Schaeffer et al. (2023) show how discontinuous metrics can change the appearance of performance trends. Accuracy thresholds our margin at zero, so we report both scales while retaining their separate noise properties. Low estimated seed signal relative to item noise provides a measurement limitation for accuracy in this study. Dehghani et al. (2021) show that benchmark selection can reorder methods, which concerns the composition sensitivity examined here.

Demšar (2006) and Dror et al. (2017) develop inference across datasets, while Card et al. (2020) examine statistical power and Gorman and Bedrick (2019) examine dependence on a single split. Burnell et al. (2023) advocate per-instance reporting alongside aggregate scores. Henderson et al. (2018) and Agarwal et al. (2021) address seed sensitivity and uncertainty in reinforcement learning. The authors resample runs within tasks, with an additional task-resampling variant in their Appendix A.5, while our application requires preservation of paired run effects across benchmarks.

Hardy et al. (2026) apply generalizability theory to 4,000 released models, which places replication across models rather than training seeds. Kohli (2026) estimates roughly two independent votes among nine frontier judges, and Messing (2026) reports naive standard errors that are 40 to 60 percent too small after accounting for judge and prompt variation. Sha and Zhao (2026) estimate how many independent signals a leaderboard’s benchmark scores provide across models. Those studies measure dependence across models, judges, or prompts, while we measure paired benchmark deviations across training runs.

The cross-half identity builds on the repeated-measurement approach to attenuation described by Spearman (1904). Under independent errors between halves, paired measurements remove the item-noise contribution from covariance estimates, including diagonal entries. Cronbach (1951) relates

split-half reliability to random halves of an item population, which motivates the reliability table and the 50 re-splits in N3. Those re-splits change the margin estimate by a reported standard deviation of 0.002.

The Brown (1910) and Spearman (1910) test-length formula contains $1 + (k-1)\rho$, the same algebraic form used for Λ^2 . Its correlation concerns parallel items, whereas ours concerns benchmark run effects. Substituting the reliability means in Table 5 at $k = 10$ would produce 6.87 and 3.91, which aren’t estimates of our variance inflation.

Campbell and Fiske (1959) compare traits and measurement methods in a crossed matrix. The within-format and cross-format summaries here resemble heterotrait blocks, but no benchmark appears in multiple formats. This design therefore can’t separate benchmark content from scoring format or establish format as the cause of the association. Brennan (2001) provides the variance-component framework for these distinctions.

Hofmann et al. (2025) fit benchmark-specific item-response models to an overlapping evaluation panel, with seven authors shared with Heineman et al. (2025). Kipnis et al. (2024) estimate a factor across six benchmarks that correlates 0.94 with total score. Those ability structures describe performance differences rather than the within-configuration noise covariance estimated here. Hedges et al. (2010) estimate uncertainty for dependent effect sizes without requiring their complete covariance matrix, while our application estimates the covariance needed for an effective-count point estimate.

C SUPPLEMENTARY RESULTS

Covariance diagnostics The largest discrepancy between opposite cross-half covariance entries is 2.44 bootstrap standard errors. This maximum across multiple pairs doesn’t by itself establish leakage. The reduced item records provide 57 MMLU subject groups but only one group for each other trait, so the passage-aware split in Appendix A uses the release’s request files instead.

Among the eight accuracy traits with positive estimated seed variance, estimated correlations range from -0.216 to 1.296 , with the maximum occurring for ARC-Easy and MMLU. The symmetrised matrix has a negative eigenvalue of -0.55 and a leading eigenvalue of 3.85 . These estimates don’t form a valid correlation matrix, a possible finite-sample outcome of noise-corrected moment estimation. We retain the unprojected matrix because a positive-semidefinite projection would change the reported values without an established correction target. The primary ratio uses cross-half sums directly and doesn’t require this correlation-matrix transformation. The nine-trait margin matrix has correlations from -0.157 to 0.802 and nine positive eigenvalues. Its leading eigenvalue is 4.33 , with a spectral participation ratio of 3.57 . The accuracy participation ratio is 3.12 before clipping its negative eigenvalue and 3.62 after clipping. Because that accuracy matrix is indefinite, neither figure provides a validated count of independent population directions.

The original scorecard counted five failures, four passes, and two predictions that couldn’t be tested as written. Predictions 9 and 11 came from a separate analysis of the per-run reductions, rather than the original pipeline. Prediction 7 lacks its required lower confidence limit, and prediction 6 depends on a post hoc choice to compare excess inflation. Table 8 retains the numerical outcomes and states where the assessment differs from the proposed criterion.

Variation across model sizes Each size band contains 25 configurations, one per recipe. Its clusters are therefore singletons, without within-band replication of sizes inside recipes. Only the 530M interval excludes one in either phenotype, the margin exclusion disappears under the test-inversion interval of Appendix C, and the intervals don’t support a ranking of model sizes. All five margin point estimates and four accuracy point estimates exceed one, so prediction 5 passes a point-estimate interpretation and fails the stricter interval interpretation used in the original scorecard.

The lower squared-ratio bound for 1B margins falls below zero, so its square-root endpoint is unavailable. Three constructions that stay positive give the 1B margin estimate finite endpoints, and none of them excludes one. The configuration percentile bootstrap gives 0.834 to 1.643 , the cluster-robust $t(24)$ interval on the log of the squared ratio gives 0.811 to 1.628 , and a wild cluster bootstrap on that log scale gives 0.491 to 1.656 . At the full battery the log-scale wild interval is 1.145 to 1.338 , which agrees with the reported 1.143 to 1.338 at the second decimal, so the transform matters

Table 8. Exploratory scorecard for the eleven reported planning predictions. Several outcomes depend on interpretations absent from the planning document, and the table records those limitations.

No.	Criterion	Measured result	Assessment
1	Margin above 1.349, accuracy above 1.40, lower limits above 1	1.244 and 1.078	Neither point threshold reached
2	Accuracy-to-margin ratio at least 0.8	0.867	Point criterion met
3	Retained excess above 0.70 and N5 share below 0.30	1.23 and 0.61	Default N5 criterion fails
4	N6 share below 0.50	0.015	Point criterion met
5	Underprediction in at least four bands	Intervals exclude 1 in one band	Fails interval reading, meets point reading
6	Transport within 20 percent, independence error above 20 percent	+66%, -28.9%, three units	Excess-scale criterion fails, raw scale differs
7	External estimate and lower limit above 1	1.705 for likelihood	Required interval unavailable
8	At least eight ratios in [1.4, 3.0]	Ten, range [1.571, 1.732]	Lower threshold can't discriminate
9	Cross-format diagnostic above 1.2, lower limit above 1	0.921 and 0.965	Neither criterion met
10	Frobenius distance above 0.5	5.59 and 6.49	Criterion met with fill convention
11	More than ten percent of pairs with predicted flip probability above 0.05	455 of 1,500, or 30.3 percent	Observed-gap calculation meets criterion

Table 9. Inflation estimates by model size across the 25 recipes. The intervals use the wild cluster bootstrap-t calculation described in Appendix D.

Size	Λ_M	95% interval	Λ_A	95% interval
150M	1.101	[0.781, 1.344]	1.028	[0.878, 1.165]
300M	1.350	[0.638, 1.803]	1.080	[0.836, 1.288]
530M	1.514	[1.064, 1.837]	1.219	[1.083, 1.338]
750M	1.110	[0.686, 1.407]	0.987	[0.310, 1.320]
1B	1.149	[n/a, 1.751]	1.205	[0.805, 1.497]
Pooled, $N = 125$	1.244	[1.143, 1.338]	1.078	[0.993, 1.157]

only where the squared ratio is poorly determined. A separate run scored those constructions where they actually get used, inside one size band at the three runs the release ships, and the transform doesn't repair that case. On a margin-like band the plain interval covers 0.930 and the log one covers 0.902, and at ten runs the two sit at 0.947 and 0.944. On an accuracy-like band at three runs the plain interval is undefined in 0.284 of replicates and misses in 0.017 of them, while the log interval is undefined in 0.090 and misses in 0.151, so it buys definedness by paying coverage. The leverage correction is the better repair of the two, since it lifts the margin band to 0.942 at three runs and the accuracy band to 0.952 at ten. The 1B log-scale endpoints should therefore be read as closer to a ninety percent statement than a ninety five percent one, which widens the caveat that row already carries, and that run used seed 20260972 at 4,000 replicates. Simulation says these per-size rows should be read as estimates with wide uncertainty rather than as intervals carrying their nominal level, and the accuracy column is the weaker of the two. One band of 25 configurations at the three runs the release ships covers 0.930 on a margin-like population and only 0.707 on an accuracy-like one, where the louder item noise leaves 25 clusters too little to work with. Six runs repair the accuracy case to 0.937 and ten reach 0.947, so the shortfall is a sample size problem rather than a flaw in the interval. The power to detect anything inside a single band is worse again than the coverage. At a true margin ratio of 1.2446 a single band excludes one in 0.313 of replicates at three runs, 0.739 at six and 0.957 at ten, while at a true accuracy ratio of 1.078 the exclusion rate reaches only 0.081 at ten runs and 0.317 at forty. No affordable run count settles the accuracy question inside

one band, which is the honest reason the paper draws its conclusions from the pooled design. These ran at 4,000 replicates with seeds 20260963 and 20260971. Figure 2 reports N1 calibration at six effect sizes with 2,000 replicates each, using the 2.5th and 97.5th percentiles of estimates as dispersion bands. Those bands describe simulated estimator dispersion rather than confidence-interval coverage.

Permutation comparison N2 was proposed as an estimator-bias check rather than the primary inferential procedure. In the reported 500 permutations, none reaches the observed margin estimate, and 0.028 reach the accuracy estimate. This reassignment holds the observed configurations fixed while independently permuting trait-specific run labels. The primary bootstrap instead targets variation across recipe clusters, so the two procedures use different randomisation and sampling assumptions. We report both accuracy results because the disagreement depends on their different randomisation and sampling assumptions.

Associations with scoring format The cross-format diagnostic retains the diagonal sum and only those off-diagonal terms whose traits have different formats. It has a null reference of one when those retained covariances vanish, but the masked sum needn't represent the variance of a realisable benchmark average. Because each benchmark has only one format, the comparison can't identify a causal effect of format.

The separate calculation reproduces the full estimate to four decimal places before retaining 54 of the 90 ordered off-diagonal pairs. It gives 0.921 with interval 0.801 to 1.035 for margins, and 0.965 with interval 0.848 to 1.063 for accuracy. Both fall below the proposed 1.2 threshold, and the margin result falls below the 1.153 expected from uniformly distributed covariance contributions under the comparison used here.

BoolQ is the only yes/no benchmark, so all its off-diagonal pairs cross a format boundary. Its margin seed variance is 1.43×10^{-3} , compared with 4.34×10^{-4} for the next-largest trait. Excluding BoolQ gives a cross-format margin diagnostic of 1.006 with interval 0.972 to 1.039, compared with an unrestricted nine-trait estimate of 1.786 with interval 1.689 to 1.877.

Among the nine traits with defined margin correlations, excluding WinoGrande, the mean correlation is 0.624 across sixteen within-format pairs and 0.007 across twenty cross-format pairs. The three cloze traits give margin inflation of 1.225 with interval 1.064 to 1.367. Excluding ARC-Easy and ARC-Challenge from the multiple-choice block leaves inflation of 1.479 with interval 1.416 to 1.541. These associations show concentration within the larger format groups, while benchmark content and format remain confounded. A rank-one model with trait-specific loadings can also produce uneven contributions across those groups.

Scoring-scale interventions We transformed the saved margins item by item and recomputed each estimate with the items, runs, split, and interval routines held fixed. For benchmark j , s_j is the standard deviation of its per-item margins pooled across all configurations, runs, and items, and it ranges from 0.087 for WinoGrande to 0.468 for CommonsenseQA. Table 10 reports the untransformed margins, division by s_j , clipping at $\pm s_j$ before that division, two hyperbolic tangent scores, and the sign of each margin. Multiplying every margin by ten leaves the estimate unchanged to nine decimal places, which checks the construction.

Division by s_j changes only the benchmark weights, and it lowers margin inflation from 1.244 to 1.182. The three bounded scores lower it further, to between 1.120 and 1.141, and the sign score gives 1.079, which differs from the accuracy estimate of 1.078 by 0.0007 for reasons we haven't traced. Across these scores the off-diagonal sum falls from 0.547 to 0.164 times the trace, while BoolQ's share of that trace grows from 0.679 to 0.844. Bounding the margins of the same runs therefore reproduces between 62 and 75 percent of the gap between margin and accuracy inflation. These transformations change the scoring rule and leave the trained models untouched, so they don't show how training produces large margins, and the intervals treat each s_j as fixed.

Table 11 applies $\bar{r}_E = 0.0181$ while holding trait standard deviations fixed. Its adjusted column applies to the weighting of this battery, with flip probabilities interpreted under the stated Gaussian comparison model. The held-out accuracy point errors favour the full matrix over independence, 0.0105 versus 0.0143, without an interval for their difference. Every predictor also receives held-out

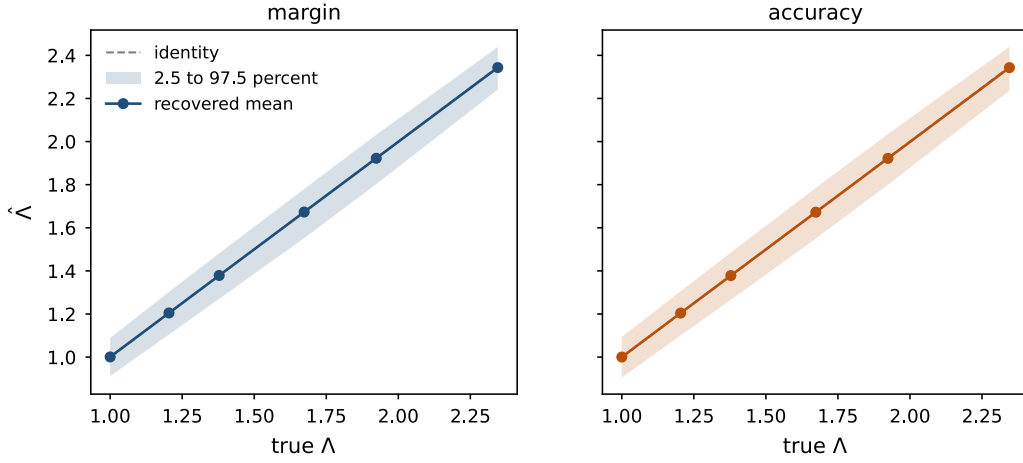


Figure 2. Reported simulation recovery across six specified covariance settings for both score scales. The shaded bands describe replicate dispersion and don’t establish interval coverage under other data-generating assumptions.

Table 10. Margin inflation after item-level transformations of the same saved margins, with wild recipe-cluster intervals over 4,999 draws. s_j is the pooled standard deviation of benchmark j ’s item margins.

Item score	$\hat{\Lambda}$	Interval	Off-diagonal sum / trace	BoolQ trace share
m	1.244	[1.143, 1.338]	0.547	0.679
$10m$	1.244	[1.143, 1.338]	0.547	0.679
m/s_j	1.182	[1.094, 1.265]	0.398	0.786
$\min(\max(m, -s_j), s_j)/s_j$	1.134	[1.048, 1.213]	0.287	0.827
$\tanh(m/s_j)$	1.141	[1.053, 1.222]	0.301	0.822
$\tanh(2m/s_j)$	1.120	[1.032, 1.200]	0.254	0.831
$\text{sign}(m)$	1.079	[0.992, 1.158]	0.164	0.844

Table 11. Accuracy-point calculations at the median 300M configuration. The four- and seven-trait rows evaluate the same ten-trait quantities at different K values and aren’t additional measured batteries.

K	Independent σ_{agg}	Adjusted σ_{agg}	K_{eff}	P_{flip} , one-point gap
4	0.754	0.774	3.79	0.181
7	0.570	0.600	6.31	0.119
10	0.477	0.514	8.60	0.085

seed standard deviations unavailable without replication, so these results don’t establish an operational replacement for replicated runs. The three-unit PolyPythias accuracy estimate of 2.286 differs from 1.078 and has a configuration-bootstrap interval with no defined lower endpoint, which further limits transport of this calculation.

Performance correlation as a noise proxy The reported Frobenius distances between R_E and R_P are 5.59 for margins and 6.49 for accuracy. Negative estimated variances leave eighteen margin and 34 accuracy ordered off-diagonal correlations undefined. The implementation fills these entries with zero before differencing, including undefined diagonal entries compared with unit diagonals in R_P . Those diagonal substitutions contribute 1.0 and 2.0 to the squared distances.

Repairing only those diagonals gives distances of 5.499 and 6.339. Restricting the comparison to jointly defined entries gives 4.374 and 4.593, which still exceed the planned threshold of 0.5. The

Table 12. Reported null and sensitivity results with replicate standard deviations. N3 and N4 use the observed correlation and therefore describe sensitivity rather than a zero-correlation null.

Test	Structure	$\hat{\Lambda}_M$	SD	$\hat{\Lambda}_A$	SD
N1	Zero correlation with observed counts and reliabilities	1.0009	0.046	1.0004	0.048
N2	Independent trait-specific run-label permutations	1.0010	0.048	0.9984	0.039
N1	$\bar{r}_E = 0.05$, true $\Lambda = 1.2042$	1.2048	0.051	1.2044	0.053
N1	$\bar{r}_E = 0.10$, true $\Lambda = 1.3784$	1.3788	0.055	1.3782	0.057
N1	$\bar{r}_E = 0.20$, true $\Lambda = 1.6733$	1.6730	0.059	1.6724	0.061
N1	$\bar{r}_E = 0.30$, true $\Lambda = 1.9235$	1.9226	0.059	1.9220	0.061
N1	$\bar{r}_E = 0.50$, true $\Lambda = 2.3452$	2.3433	0.052	2.3429	0.053
N3	Fifty independent item re-splits	1.2459	0.002	1.0843	0.008
N4	Zero mediation with pooled-slope regression	1.0910	0.096	1.1045	0.062
N5	Zero correlation plus gain, $\sigma_g = 0.05$	1.1477	0.050	0.9986	0.048
N6	Zero correlation plus default-versus-auxiliary offset	1.0038	0.044	1.0030	0.047
N6	Same offset with auxiliary-run contrast	1.0004	0.062	1.0003	0.065

threshold alone offers limited discrimination between different ten-trait matrices, so the entrywise comparisons provide a more useful description.

The off-diagonal association between the matrices is 0.756 for margins and 0.345 for accuracy. Mean defined R_E entries are 0.281 and 0.273, compared with all-pair R_P means of 0.750 and 0.754. On matching pairs, the R_P means are 0.743 and 0.769. The all-pair comparison therefore differs in scale by factors of approximately 2.7 and 2.8. The unadjusted proxy predicts a mean margin aggregate standard deviation of 0.0072, compared with the reported 0.0057.

An additional exploratory baseline, P2 R, rescales R_P off-diagonals by the training-fold ratio of mean noise correlation to mean performance correlation. Its mean factors are 0.366 for margins and 0.370 for accuracy, compared with whole-population ratios of 0.375 and 0.362. On the same five recipe folds, its squared prediction errors are 0.0131 and 0.0342. It improves on P2 but remains 2.3 times worse than P1 on margins and worse than independence on accuracy. Its direction relative to independence holds on margins in all 1,000 random fold partitions and on accuracy in 987 of them, but the recipe-bootstrap intervals below include zero on both scales. This rescaling also requires replicated runs to estimate its factor.

Paired uncertainty for the prediction comparison We reran the five-fold comparison with every prediction, target, and squared log error saved, and the rerun reproduces all twelve fold means in Table 2 exactly, with P2 R factors of 0.366 and 0.370. Before computing any contrast, we fixed eight paired differences and a practically meaningful difference of 0.005 in mean squared log error, which corresponds to a 7 percent error in the aggregate standard deviation. For intervals, we resampled the 25 recipes 2,000 times, assigned folds over the distinct original recipes so that copies of one recipe never sit on both sides of a split, and refitted the whole comparison in each draw. These intervals treat the designed recipes as exchangeable clusters, and they also absorb variation from the choice of fold partition.

On margins, P1, P1 G, and P3 each lower the error relative to independence by about 0.03, with intervals of -0.058 to -0.001 , -0.059 to -0.002 , and -0.060 to -0.004 , and between 95 and 97 percent of draws put the reduction beyond 0.005. P2 R lowers the error by 0.023, but its interval of -0.059 to 0.010 includes zero. The P1 minus P3 interval runs from -0.003 to 0.012, so these data don’t separate the rank-one model from the full matrix. On accuracy, each of the four intervals for a difference from independence includes zero, and the full matrix minus independence runs from -0.017 to 0.023.

Across 1,000 random five-fold partitions of the original recipes, independence never has the lowest margin error, while P1, P1 G, and P3 have the lowest error in 27, 33, and 33 percent of partitions. On accuracy, P3 has the lowest error in 55 percent of partitions and independence in 39 percent. The partition behind the table understates the margin advantage of the correlation models relative to a typical partition, since its independence error of 0.0358 lies below the partition median of 0.0438

and its P1 error of 0.0057 lies above the median of 0.0031. These contrasts concern correlation prediction with held-out marginal variances supplied, so they don't carry over to the operational comparison in the main text, where no model separates from independence.

We also ran the comparison on simulated populations with known seed covariance, using 300 replicates for each of the three structures in Table 17, and scored every model against both the noisy fold target and the true aggregate standard deviation. The noisy target itself has a mean squared log error against the truth of 0.011 to 0.012 on margins and 0.024 to 0.032 on accuracy. With the matched margin covariance, P1, P1 G, and P3 beat independence by 0.043 to 0.046 against the noisy target and by 0.048 to 0.051 against the truth, and P3 has the lowest true error, 0.0027 below P1. With independent seed deviations, independence has the lowest margin error under both targets, and P1, P1 G, and P3 lose between 0.001 and 0.007. On accuracy with the matched covariance, the true errors of P0, P1, P1 G, and P3 differ by at most 0.0032, which is small beside the target noise, so a five-fold accuracy comparison shouldn't be expected to separate them. These populations share one covariance across recipes and contain little systematic variation between configurations, so they don't reproduce how the performance-correlation plug-ins behave, and we don't use them to assess P2 or P2 R.

Recipe, size, and trait sensitivity Removing each recipe in turn gives margin estimates from 1.207 to 1.256 and accuracy estimates from 1.065 to 1.099. The maximum changes from the complete battery are 0.037 and 0.021. Removing each size gives ranges of 1.178 to 1.289 and 1.047 to 1.115, compared with individual-band ranges of 1.101 to 1.514 and 0.987 to 1.219. These overlapping intervals don't establish a size trend in the effect.

Trait removal changes the estimates more than recipe or size removal. Excluding BoolQ gives 1.786 with interval 1.689 to 1.877 for margins, and 1.558 with interval 1.412 to 1.687 for accuracy. Excluding CommonsenseQA gives the smallest point estimates, 1.096 and 1.014. BoolQ contributes 68 percent of the estimated margin variance trace and 84 percent of the accuracy trace, and the removal estimates imply that the off-diagonal terms involving BoolQ are negative, and they sum to about -0.15 of the margin trace and -0.06 of the accuracy trace. Its removal therefore raises the ratio through both terms of the excess ratio. The same removal lowers the aggregate standard deviation from 5.72×10^{-3} to 5.16×10^{-3} on margins and from 5.14×10^{-3} to 3.26×10^{-3} on accuracy, so a larger inflation factor here accompanies a less variable average. Removing WinoGrande is the only other removal whose accuracy interval, 1.023 to 1.178, excludes one, and Table 13 gives the remaining intervals and standard deviations. Removal is a switch, so we also turned it into a dial, lowering BoolQ's weight below the flat tenth in seven steps down to a twentieth while raising the other nine weights together to keep the battery summing to one. Margin inflation climbs from 1.2439 at the flat weighting through 1.3374 at eight percent to 1.5030 at the smallest weight we tried, and every interval along the way excludes one. Accuracy climbs from 1.0784 to 1.2311 over the same steps, and its interval stops including one at a BoolQ weight of 0.0867, which is a reduction of about one seventh. The accuracy conclusion is therefore not a knife edge in the estimate, which moves smoothly, and it is a knife edge in the weighting, since a battery that discounts BoolQ by a seventh reports dependence on accuracy as well. We report the flat weighting because the release gives no basis for any other, and a reader who has such a basis should expect the accuracy answer to move first.

The variance-share participation ratios are 1.97 for margins and 1.42 for accuracy on traits with defined seed variance. The first two ratios measure how strongly a few traits dominate variance, while the noise-correlation participation ratio of 3.57 measures concentration across matrix directions.

Standardising traits by estimated seed standard deviation gives 1.803 on nine margin traits and 1.707 on eight accuracy traits. These point diagnostics omit traits with nonpositive variance estimates and divide by uncertain standard deviations, with seven accuracy traits at or below item noise. Holding those estimated weights fixed, the wild recipe-cluster intervals are 1.544 to 2.023 and 1.228 to 2.076. These intervals omit the uncertainty in the weights themselves, so we don't score them against the planning thresholds.

Removing defined traits from the standardised batteries gives margin estimates from 1.625 to 1.910 and accuracy estimates from 1.388 to 1.780. BoolQ removal gives 1.910 and 1.761 under this standardisation. Selecting traits by positive estimated variance can change the target, and it could

Table 13. Trait-removal estimates across all 125 configurations in both phenotypes, with wild recipe-cluster intervals over 4,999 draws. Each removal row uses the nine remaining traits with equal weights of one ninth, and the final row uses all ten traits. The columns σ_{agg} and σ_{ind} give the aggregate and independence standard deviations in units of 10^{-3} , so the weighted trace equals σ_{ind}^2 and the weighted off-diagonal sum equals $\sigma_{\text{agg}}^2 - \sigma_{\text{ind}}^2$.

Removed trait	Λ_M	Interval	σ_{agg}	σ_{ind}	Λ_A	Interval	σ_{agg}	σ_{ind}
BoolQ	1.786	[1.689, 1.877]	5.16	2.89	1.558	[1.412, 1.687]	3.26	2.09
WinoGrande	1.246	[1.144, 1.340]	6.36	5.11	1.102	[1.023, 1.178]	5.87	5.32
HellaSwag	1.244	[1.142, 1.338]	6.35	5.10	1.076	[0.992, 1.153]	5.69	5.29
PIQA	1.243	[1.139, 1.340]	6.35	5.10	1.073	[0.988, 1.150]	5.69	5.30
MMLU	1.205	[1.103, 1.300]	6.13	5.09	1.068	[0.984, 1.143]	5.66	5.30
ARC-Challenge	1.195	[1.113, 1.272]	6.08	5.09	1.039	[0.952, 1.116]	5.47	5.27
OpenBookQA	1.178	[1.092, 1.257]	5.97	5.07	1.074	[0.992, 1.150]	5.66	5.27
ARC-Easy	1.161	[1.089, 1.228]	5.81	5.01	1.046	[0.957, 1.126]	5.43	5.19
Social IQa	1.139	[1.068, 1.205]	5.68	4.99	1.032	[0.950, 1.109]	5.41	5.24
CommonsenseQA	1.096	[1.054, 1.137]	4.99	4.55	1.014	[0.936, 1.086]	5.15	5.08
None	1.244	[1.143, 1.338]	5.72	4.59	1.078	[0.993, 1.157]	5.14	4.77

raise the mean correlation if excluded low-signal traits also have low true correlations. We haven’t tested that additional assumption, and subset stability doesn’t resolve uncertain denominators.

The raw trait-removal spans are 0.690 for margins and 0.544 for accuracy, compared with interval widths of 0.195 and 0.164. The resulting ratios of 3.5 and 3.3 compare different measures of spread and aren’t uncertainty decompositions. Applying the delete-one-trait jackknife formula produces scales of 0.554 and 0.455, compared with sampling standard errors of 0.045 and 0.038. Their ratios are 12.3 and 12.1, partly because the jackknife multiplier is 2.85. Comparing the unscaled dispersion instead gives ratios of 4.33 and 4.24.

Omitting the BoolQ removal value from that calculation gives scales of 0.139 and 0.073, or 3.08 and 1.93 times the reported sampling standard errors. These are descriptive composition diagnostics, because the ten fixed benchmarks aren’t established exchangeable draws from a benchmark population. Adding the original jackknife scales and sampling standard errors in quadrature gives 0.556 and 0.457. A normal calculation then gives ranges of 0.15 to 2.33 and 0.18 to 1.974, both including one. Repeating that arithmetic without the BoolQ removal value gives 0.958 to 1.529 and 0.918 to 1.239.

Those calculations aren’t validated confidence intervals for a benchmark population. They can double-count sampling variation, and their normal approximation behaves poorly near the parameter boundary. For example, $\Lambda = 0.15$ corresponds to $\bar{r}_E = -0.109$, only approximately 0.003 above $-1/9$. We retain the fixed-battery target and report composition sensitivity without treating these scales as an identified decomposition of total uncertainty.

Changing trait count also changes Λ at fixed effective covariance coefficient. At $\bar{r}_E = 0.061$, the nine-trait reference is 1.219 rather than the ten-trait 1.244, while the corresponding accuracy values are 1.070 and 1.078. Four margin removals and five accuracy removals exceed those mechanical references. On eight traits, standardised accuracy inflation of 1.707 implies 2.75 effective benchmarks, while the raw accuracy effective coefficient of 0.0181 implies 7.10 effective benchmarks. Comparing 2.75 with 7.10 holds trait count fixed, unlike comparison with the ten-trait value of 8.60.

The ARC pair has an estimated margin correlation of 0.801. Removing both benchmarks gives margin inflation of 1.118 with interval 1.059 to 1.174, and accuracy inflation of 1.012 with interval 0.929 to 1.086. The pair affects the estimate, although the remaining margin interval still excludes one.

Benchmark count, identity, and weights To separate benchmark count from benchmark identity, we computed both estimates for all 1,013 equal-weight subsets of two to ten benchmarks, each with a wild recipe-cluster interval over 4,999 draws. Table 14 summarises these subsets separately for each size K . Median margin inflation lies between 1.012 and 1.244 across sizes, and the share of

subsets whose interval lies above one rises from 38 percent at two benchmarks to 98 percent at eight. Accuracy medians stay between 1.017 and 1.078, and at each size from two to nine benchmarks between 20 and 31 percent of accuracy intervals lie above one. Among subsets of two to five benchmarks, 16 accuracy subsets have a nonpositive independence variance or a negative estimate of squared inflation, and so no defined estimate, and the table’s quantiles omit them. Subsets move one weight to zero and leave the rest flat, which probes only the corners of the weighting, so we also rebuilt the statistic under arbitrary weights and swept the interior. Rebuilding the numerator and the denominator from the same contrast projections reproduces the shipped values to $1e-19$ under flat weights, which is the check that licenses the sweep. Over 2,000 weightings drawn from a flat Dirichlet, 0.817 of margin intervals still exclude one, and the estimate runs from 0.976 to 1.991 with a 5th percentile of 0.999. Over 2,000 weightings drawn from a Dirichlet concentrated near the flat point, every margin interval excludes one and the lowest estimate is 1.059. Accuracy behaves in the direction that favours the reported claim, because 0.386 of flat-Dirichlet intervals and 0.402 of near-flat ones exclude one while the flat weighting itself gives 0.993 to 1.157 and doesn’t. The weighting we shipped is therefore the conservative one on accuracy, and the margin conclusion holds across four fifths of the simplex and all of its neighbourhood.

Holding the number of benchmarks fixed, subsets that contain BoolQ have mean margin inflation lower by 0.115 at two benchmarks and by 0.596 at nine, and the corresponding accuracy differences are 0.076 and 0.499. At nine benchmarks every other benchmark raises mean margin inflation, by amounts from 0.004 for WinoGrande to 0.171 for CommonsenseQA. Pooled across sizes, margin estimates have a median of 1.136 with quartiles 1.018 and 1.390, and they run from 0.927 for HellaSwag with WinoGrande up to 1.786 in subsets that drop BoolQ and WinoGrande. Accuracy estimates have a median of 1.038 with quartiles 1.001 and 1.229, and they run from 0.315 to 5.439. Two thirds of margin intervals exclude one, always from above, against 0.305 of accuracy intervals. A separate implementation recomputed every subset and reproduced each size’s median and share, together with the full-battery 1.244 with 1.143 to 1.338 and 1.078 with 0.993 to 1.157. The subsets share runs, items, and configurations, so their spread describes this battery and doesn’t sample a wider population of batteries.

Weighting by item count changes the weights of all ten benchmarks without selecting on scores, and it gives margin inflation of 1.119 with interval 1.068 to 1.170 and accuracy inflation of 1.043 with interval 0.963 to 1.119, while MMLU and HellaSwag carry 64 percent of the weight. A second reweighting, by the inverse pooled seed standard deviation, reproduces the standardised diagnostic above, which has to drop the benchmarks with nonpositive variance estimates.

Table 14. Equal-weight subsets of the ten benchmarks, grouped by size K . Quantiles cover subsets with a defined estimate, and the share above one counts subsets whose wild recipe-cluster interval lies entirely above one.

K	Subsets	Margin				Accuracy			
		Median	10th pct.	90th pct.	Above one	Median	10th pct.	90th pct.	Above one
2	45	1.012	0.988	1.267	38%	1.017	0.918	1.226	24%
3	120	1.137	0.987	1.368	54%	1.020	0.914	1.305	29%
4	210	1.210	0.986	1.467	59%	1.030	0.958	1.344	29%
5	252	1.138	0.990	1.562	62%	1.035	0.973	1.398	31%
6	210	1.103	1.009	1.607	70%	1.041	0.979	1.437	30%
7	120	1.131	1.034	1.675	86%	1.042	0.988	1.467	31%
8	45	1.160	1.066	1.692	98%	1.046	1.007	1.497	29%
9	10	1.200	1.135	1.300	100%	1.071	1.031	1.148	20%
10	1	1.244	1.244	1.244	100%	1.078	1.078	1.078	0%

Interval construction and cluster sensitivity A percentile bootstrap over 125 configurations gives intervals of 1.069 to 1.426 for margins and 0.976 to 1.182 for accuracy. The margin interval excludes one, and the accuracy interval includes one. Comparing squared linearised standard errors, using 0.045 against 0.094 on margins and 0.038 against 0.052 on accuracy, gives ratios of 0.23 and 0.53. The implementation separately reports numerator-based design effects (Kish, 1965) of 2.06 and 1.26 from $1 + 4\rho_{ICC}$, compared with the planned 1.60.

These ratios describe different quantities and aren't contradictory estimates of one design effect. The numerator calculation uses T_c , while linearisation uses $T_c - \hat{\Lambda}^2 U_c$. The interval procedures also change resampling unit, studentisation, and denominator treatment together. We consequently don't infer a single intraclass correlation from their relative widths. The bootstrap-t addresses concerns associated with few-cluster approximations, as discussed by Bell and McCaffrey (2002), but it doesn't guarantee coverage at 25 clusters.

Observed gaps and predicted rank reversals Using accuracy noise $\sigma_{\text{agg}} = 0.00514$, the Gaussian calculation gives $P_{\text{flip}} > 0.05$ for a true gap below 0.01196. Among the $5\binom{25}{2} = 1,500$ observed same-size pairs, 455 have observed gaps below this threshold, or 30.3 percent. The proportions range from 28.7 to 33.3 percent across bands. Substituting observed gaps meets the stated numerical criterion, but noisy observed gaps can move pairs across the threshold in either direction. This count therefore isn't a lower bound on the fraction of true gaps below the threshold, and it doesn't directly measure realised reversal frequency. A practitioner faces the operational form of the same question, which is whether correcting their own test for this covariance helps once the correction carries the noise of a three-run design. We simulated that question directly rather than arguing it. Each of 4,000 replicates builds the shipped design, estimates the ratio from it, and then applies three rules to a fresh comparison of two model averages, which are an uncorrected rule that assumes benchmark independence, a plug-in rule that multiplies its standard error by the estimate, and an oracle rule that uses the true value. On a margin-like population the uncorrected rule declares a difference in 0.113 of replicates when the true gap is zero, against its nominal 0.05, while the plug-in rule declares in 0.051 and the oracle in 0.050. Three runs per model leave those at 0.115, 0.053 and 0.051. The estimate carries a standard deviation of 0.059 across replicates and still lands within 0.001 of the oracle, so its own noise costs almost nothing at this design. On an accuracy-like population the gap is smaller and the estimate noisier, at a standard deviation of 0.099, and there the three rules give 0.067, 0.052 and 0.047. The plug-in rule overshoots the oracle by 0.005 there, which is the price of estimating a correction that is itself close to one. A practitioner who accepts that correction will ask next how many runs their own comparison needs, and whether buying runs also repairs the miscalibration, so we swept the same three rules over one to ten runs per model at true gaps of zero, a quarter, a half and one aggregate standard deviation, with 4,000 replicates per combination. Extra runs buy nothing against the size error, because the true standard error and the independence standard error both fall with the square root of the run count and their ratio doesn't move at all. The sweep says so directly, since the uncorrected rule declares at a zero margin gap in between 0.112 and 0.122 of replicates across all ten run counts, while the corrected rule stays between 0.047 and 0.056, and the accuracy population gives 0.061 to 0.078 against 0.047 to 0.063. Neither sequence trends with the run budget, and each individual rate carries a Monte Carlo standard error near 0.004, so what looks like movement is the simulation rather than the design. Power is the thing that additional runs actually buy here. At a gap of one aggregate standard deviation the corrected rule declares in 0.100 of replicates with one run per model and in 0.614 with ten, while the accuracy population gives 0.111 and 0.615 at the same two budgets. No run count up to ten reaches four fifths at any of the three nonzero gaps, so a calibrated comparison at this battery costs more runs than the release supplies, and the correction fixes the error rate rather than the sample size. A second sweep carried the same design out to 80 runs per model and found the size error exactly where it had been, since across 12, 16, 20, 24, 32, 40, 60 and 80 runs the uncorrected rule declares at a zero margin gap in between 0.108 and 0.119 of replicates while the corrected rule holds between 0.047 and 0.053. Four fifths arrives at 16 runs per model for a margin gap of one aggregate standard deviation, which is exactly what the ordinary normal calculation for that gap gives, and at 20 runs on the accuracy population, where the oracle rule also misses four fifths at 16 runs by 0.002. The extra accuracy runs therefore pay for the normal approximation rather than for the estimated correction. A gap of half a standard deviation needs 63 runs per model by the same calculation, and the nearest count above it that we simulated, 80, reaches 0.881 on margins. A third sweep took the quarter standard deviation gap out to 360 runs per model and found four fifths at exactly the 252 runs the closed form predicts for margins, and at 300 on the accuracy population, where even the oracle rule falls a fraction short at 252. The uncorrected rule still declares at a zero margin gap in between 0.112 and 0.122 of replicates at every one of those seven counts, so the size error survives every run budget from one run per model to 360. Those are the numbers a replicate study should be sized against, and all of them sit far above the three runs per configuration that this release ships. Both sweeps estimate the correction on the same population that the comparison is drawn from, which is

the friendly case and not the one a reader is in. Anyone applying this correction will estimate the ratio once, on one battery and one model family, and will then carry it to a comparison whose own dependence they haven't measured, so we measured that transfer directly. The correction was estimated from the shipped margin design, whose true ratio is 1.245, and it was applied to comparisons whose own true ratio was swept from 1.00 to 1.60, with 8,000 replicates per target and a matched rule re-estimated on the target for comparison. The transferred rule holds nominal size only where the two ratios agree, at 0.053 when the target is 1.245 itself. Below that it turns conservative rather than wrong, declaring in 0.015 of replicates at a target of 1.00 and 0.028 at 1.10, and it pays for that in power, since a gap of one aggregate standard deviation at a target of 1.00 is declared in 0.043 of replicates against 0.114 for the matched rule. Above it the rule under-corrects and the size error comes back, at 0.080 for a target of 1.40, 0.101 at 1.50 and 0.123 at 1.60, against 0.159, 0.188 and 0.216 for no correction at all. The matched rule holds between 0.048 and 0.053 across the entire sweep. The correction is therefore not a constant that a reader can borrow from this paper. What transfers is the procedure rather than the number, and a practitioner whose battery is more dependent than ours would keep most of the error they were trying to remove. Repeating the whole sweep with the accuracy design as the source, whose true ratio is 1.078 and whose estimate carries a standard deviation of 0.098, gives the same shape from the other end. That correction holds nominal size at 0.056 for a target of 1.10, which is its own neighbourhood, and it recovers almost nothing above that, at 0.168 for a target of 1.50 against 0.199 with no correction at all and 0.054 for a matched rule. The penalty is therefore governed by the distance between the two ratios rather than by which design produced the number. Reading both sweeps together, the transferred rule keeps its declare rate between 0.04 and 0.06 only while the target ratio sits within about 0.05 of the source, which is a narrow enough window that we recommend measuring rather than borrowing.

Adjacent-checkpoint variation We compare all nine 160M PolyPythias runs at steps 143000 and 142000 on a nested 1,800-item sample. The check includes nine non-MMLU traits and fixes items, scoring code, and batching. MMLU's 57 subjects would increase the forward passes approximately threefold to include one additional trait, so this check excludes it.

The margin battery's between-run standard deviation is 0.00513, while the root-mean-square checkpoint shift is 0.00456, a ratio of 0.890. Accuracy gives 0.00451 and 0.00710, a ratio of 1.573. Across the nine runs, final and preceding checkpoint scores correlate 0.573 for margins and -0.070 for accuracy. Our repeat rescore reproduces each of these values. Accuracy rises by 0.0041 on average between the two steps, and removing that shared change lowers the accuracy ratio to 1.279, while the margin ratio stays at 0.890. These results show substantial checkpoint sensitivity in this restricted sample, and the design can't fully separate stochastic trajectory variation from systematic progress across 1,000 steps.

The estimator describes run variation at the scored checkpoint, which can include trajectory effects correlated across traits. It doesn't isolate initialisation from these effects in the available design. This comparison uses 1,800 rather than 4,755 items, nine runs, and a final checkpoint, whereas DataDecide uses common steps during training. On margins, BoolQ alone has between-run standard deviation 0.0438 and root-mean-square shift 0.0370 in this check, and we didn't compute these values for the other benchmarks. The next paragraph measures the corresponding checkpoint sensitivity in DataDecide.

Adjacent-checkpoint variation in DataDecide We reduced every DataDecide run a second time at the largest checkpoint step that all three runs in its configuration share below the selected step. The gap is 1,250 steps at 530M and 750M, between 657 and 1,250 steps at 150M and 300M, and 1,869 steps at 1B. In two configurations, dclm-baseline-75p-dolma1.7-25p at 750M and dolma1.7-no-math-no-code at 1B, the three runs share no released step below the selected one, so the comparison covers 369 runs in 123 configurations. Table 15 reports both estimates at the two steps with the same items, split, weights, and interval routines.

Margin inflation falls from 1.233 to 1.192, and a paired recipe bootstrap over 4,999 draws gives a 95 percent interval of -0.078 to 0.001 for the change. Accuracy inflation rises from 1.082 to 1.130, with a change interval of -0.022 to 0.130. Neither change interval excludes zero, but the accuracy interval at the earlier step, $[1.044, 1.210]$, excludes one, so the reading of the full-battery accuracy estimate as close to independence depends on the scored checkpoint. A rerun with the seed-noise estimator on the same 123 configurations reproduces all four estimates, and the centred

test-inversion interval of Appendix C gives 1.059 to 1.222 for accuracy at the earlier step, so that exclusion of one doesn't depend on the wild construction. The same inversion gives 0.996 to 1.169 for accuracy at the selected step and lowers both margin lower endpoints, to 1.099 at the selected step and 1.080 at the earlier one. BoolQ's share of the accuracy trace moves only from 0.847 to 0.872 between the two steps.

In the form used for PolyPythias, the root-mean-square change in a run's aggregate score equals 0.469 between-seed standard deviations on margins and 0.631 on accuracy. After we remove each configuration's mean change, the ratios are 0.493 and 0.618, so the movement differs between seeds and doesn't reflect shared training progress alone. Within configurations, the two steps' seed deviations correlate 0.870 for margins and 0.811 for accuracy, and the three runs keep their order in 64 and 52 percent of configurations. The 750M band, where the selected step lies furthest from the final step, gives ratios of 0.889 and 1.048 with correlations of 0.502 and 0.511. The 750M margin ratio equals the PolyPythias value of 0.890, while the accuracy ratio stays below the PolyPythias value of 1.573, and the DataDecide gap is 1,250 steps rather than 1,000. The other four sizes give ratios between 0.220 and 0.551. We added this comparison after seeing the full-sample estimates, so it is exploratory like the rest of the empirical study.

Table 15. Inflation at the selected step and at the previous step shared by all three runs, on the 123 configurations that have both. Estimate intervals use the wild cluster bootstrap- t , and the change interval uses a paired recipe bootstrap, both over 4,999 draws. The shift ratio is the root-mean-square change in a run's aggregate score divided by the between-seed standard deviation.

Scale	Selected step	Previous step	Change	Shift ratio	At 750M
Margin	1.233 [1.139, 1.319]	1.192 [1.104, 1.275]	-0.041 [-0.078, 0.001]	0.469	0.889
Accuracy	1.082 [0.992, 1.163]	1.130 [1.044, 1.210]	0.048 [-0.022, 0.130]	0.631	1.048

Simulation and permutation calibration The linearised margin standard error of 0.045 is close to the N1 null dispersion of 0.046. Accuracy gives 0.038 against the simulated 0.048, a difference of approximately 21 percent. The reported interval half-widths are 0.0974 and 0.082, approximately 2.2 standard errors compared with the $t(24)$ critical value of 2.06. Substituting the accuracy simulation dispersion gives a descriptive interval of 0.979 to 1.178, which also includes one. This substitution isn't a calibrated alternative interval because the simulated and observed populations differ.

A separate implementation of the interval definitions uses 25 clusters of five configurations, three runs, and ten traits. It assumes exchangeable seed covariance, independent item noise across halves, and recipe heterogeneity in seed variance. Across sixteen cells formed from four true correlations between zero and 0.10 and four noise settings, 200 datasets per cell give empirical coverage from 0.920 to 0.975. Each cell lies within two Monte Carlo standard errors of 0.95, and estimated inflation is within 0.01 of truth in fifteen cells. Four cells with dispersion 0.044, compared with observed null dispersions 0.046 and 0.048, have coverage 0.950, 0.935, 0.975, and 0.960. Their interval widths range from 0.203 to 0.238, compared with the reported margin width of 0.195. This limited simulation checks the procedure under its specified populations and doesn't validate all departures present in the study.

A second simulation, run for this revision with 2,000 replicates per population and 4,999 draws, scores four interval procedures on the same simulated moments. Table 16 lists the populations, each with 125 configurations in 25 recipe clusters, three runs, and ten benchmarks. The wild bootstrap- t covers between 0.928 and 0.954. The cluster-robust $t(24)$ tracks it within 0.02 and falls to 0.911 with one recipe at six times the variance. The configuration percentile bootstrap falls to 0.872 under recipe-shared run effects, and the recipe percentile bootstrap covers between 0.857 and 0.940 with undefined draws in the accuracy-like noise cells. Cross-half error correlation of 0.1 biases the estimate by -0.014 and -0.016, and its margin-like wild interval misses above in 0.066 of replicates. The Monte Carlo standard error of a coverage near 0.95 is 0.005.

We later added a cluster test-inversion interval, a family the analysis plan lists for ratios with an uncertain denominator. For each candidate $\theta = \Lambda^2$, it forms the recipe sums $z_g = \sum_{c \in g} (T_c - \theta U_c)$ and keeps θ when the cluster-robust t statistic of their mean, with a $G/(G-1)$ factor, stays within

the $t(24)$ critical value. The accepted set can be unbounded, and we record its shape instead of truncating it. Rerunning the R6 generator with the same seeds reproduces the cluster- t coverage of every row in Table 16, so the inversion column scores the same 24,000 simulated datasets. The inversion covers between 0.932 and 0.957, with median widths within 0.006 of the wild interval, and it holds 0.955 with one recipe at six times the variance, where the wild interval gives 0.930. Its set is bounded in 23,999 of those datasets, and the exception is one Student- t replicate. A restricted variant that imposes θ in the variance without centring covers between 0.946 and 0.970, which is above 0.95 in eleven of the twelve populations and more than two Monte Carlo standard errors above it in eight.

The Monte Carlo standard error of 0.005 at 2,000 replicates is the size of the shortfalls that Table 16 shows, so we repeated four cells with 10,000 replicates and new seeds. They are the two cells calibrated to the observed estimates and the two with the lowest wild coverage. The wild interval covers 0.933 with cross-half correlation 0.1, 0.935 with one dominant recipe, 0.940 in the margin-like cell, and 0.945 in the accuracy-like cell, each with a Monte Carlo standard error near 0.0025. We later repeated these four cells a second time at 10,000 replicates with a third seed, which returns 0.937, 0.935, 0.951 and 0.947. Three of them agree with the first run inside one standard error. The margin-like cell doesn't, because 0.940 and 0.951 sit more than three standard errors of their difference apart, so we ran that one cell a third time at 40,000 replicates, which returns 0.947 with a Monte Carlo standard error of 0.0011 and puts the residual-scaled version at 0.955. Pooling all three runs of that cell over 60,000 replicates gives 0.946, so it is the mildest of the four rather than a settled 0.940. The cross-half and dominant-recipe cells repeat within 0.004, and they carry the shortfall that matters here. In the cross-half cell the wild interval misses above in 0.054 of replicates against a nominal 0.025, so the shortfall falls on one side. The centred inversion covers 0.935, 0.956, 0.942, and 0.947 in the same four cells, so the inversion keeps its advantage of 0.021 under one dominant recipe while matching the wild interval elsewhere. We then repeated the other eight cells at the same precision, which puts every cell of Table 16 on 10,000 replicates for these four procedures. No cell moves by more than 0.011 against its 2,000-replicate value. Across the twelve, the wild interval covers 0.933 to 0.953, the cluster-robust t covers 0.916 to 0.949, the centred inversion covers 0.935 to 0.956, and the restricted inversion covers 0.951 to 0.970. The restricted inversion covers at least as much as the centred one, which covers at least as much as the wild interval, which covers at least as much as the cluster-robust t , in eleven of the twelve cells. The exception is the cell with mean correlation 0.2, where the wild interval beats the centred inversion by 0.002. The three lowest wild coverages are 0.933 and 0.936 in the two cross-half correlation cells and 0.935 with one dominant recipe. A later sweep repeated all twelve populations once more at 10,000 replicates with a third seed, alongside the residual-scaled construction that the next paragraph compares, and it puts the wild interval between 0.929 and 0.950. That second sweep agrees with this one to within 0.004 at each end, so the shortfall we describe here is a property of the construction rather than of one set of draws.

Coverage in simulation says which construction a reader should trust, and it doesn't say how far the reported interval would move under a different one, so we applied all eight to the data itself. Each wild construction ran 9,999 draws over twenty bootstrap seeds, and no endpoint varies across those seeds by a standard deviation above 0.0023. That matters most for the accuracy conclusion, whose lower endpoint of 0.9933 is the one number that decides whether the full-battery interval includes one. Its standard deviation across the twenty seeds is 0.0015, so the endpoint sits more than four of its own bootstrap standard deviations below one, and the inclusion of one is a property of the data rather than of the draws. On the full battery the margin interval runs from 1.141 to 1.339 under Rademacher weights, 1.166 to 1.344 under Mammen weights, 1.142 to 1.338 under Webb weights, and 1.130 to 1.349 under leave-one-cluster-out residual scaling, with the cluster-robust t at 1.147 to 1.334, the centred inversion at 1.095 to 1.323, and the restricted inversion at 1.074 to 1.329. Every one of the eight excludes one on that scale. On accuracy every one of the eight includes one instead, and the widest of them, the restricted inversion, runs from 0.988 to 1.169. Removing BoolQ leaves both scales above one under all eight, where the widest margin interval is again the restricted inversion at 1.391 to 1.840. The shipped interval, which draws 4,999 times rather than 9,999, sits within 0.003 of the Rademacher mean in all four batteries. Which of these intervals include one, and which exclude it, doesn't depend on the construction we report.

None of that repairs the wild interval's own shortfall, so we asked whether a different construction would. We compared six of them on the same twelve populations with 4,000 replicates each, which

are the wild bootstrap- t under Rademacher, Mammen and Webb weights, a wild bootstrap- t whose recipe residual is divided by $1 - h_g$ with h_g the recipe's share of the denominator sum, the cluster-robust t with the $G/(G - 1)$ correction, and the centred inversion. We ran four cells first, which are the two with the lowest wild coverage and the two calibrated to the observed estimates, and there the residual-scaled version reached nominal coverage in three of them while no other construction reached it in more than one, so we then ran the remaining eight to see whether the advantage survived, and it did. Across all twelve at that replicate count it covers between 0.943 and 0.958, against 0.932 to 0.951 for the shipped Rademacher version. We then repeated both of those two constructions on all twelve populations at 10,000 replicates, which cuts the Monte Carlo standard error to 0.002 and puts the comparison at one precision on both sides. The residual-scaled version covers between 0.940 and 0.959 there, against 0.929 to 0.950 for the shipped one, and it improves every cell by between 0.007 and 0.014. It reaches 0.95 in ten cells where the shipped version reaches it in one, and its intervals run 4 to 13 percent wider at the median. Mammen weights cover between 0.916 and 0.942 and lose to Rademacher in every cell, Webb weights land within 0.003 of Rademacher everywhere, and the corrected cluster-robust t covers between 0.923 and 0.950. The residual-scaled version also matches or beats the centred inversion in all twelve, by up to 0.012. We picked that construction after seeing the first four cells, so the eight that followed confirm it on the same generator family rather than test it independently, and we therefore report the shipped interval throughout and record this one as the better-calibrated alternative. On the observed data it gives [1.130, 1.349] for margins and [0.987, 1.163] for accuracy, so it moves no conclusion, except that its upper endpoint erases the 0.011 of separation from the planned threshold that we report below.

A reader can also ask what the reported interval becomes once its own miscalibration is corrected on these data, rather than through a construction chosen in simulation. We applied Beran pre-pivoting, which runs a second wild bootstrap of 1,999 draws inside each of 1,999 outer resamples, and reads the outer quantiles at the levels the inner distribution implies. On the full battery the margin interval calibrates to [1.151, 1.328], which runs 0.018 narrower than the shipped [1.143, 1.338] and still excludes one. Accuracy calibrates to [0.994, 1.161] and still includes one, the BoolQ-removed margin interval calibrates to [1.689, 1.878], and the BoolQ-removed accuracy interval widens to [1.396, 1.704] while staying above one. The implied coverage of the nominal levels runs from 0.925 to 0.977 across the four batteries, so this correction moves no conclusion, and on the full battery it disagrees with the simulation about the direction of the error. Pre-pivoting measures calibration inside the resampling model that the interval already assumes, so a violation the bootstrap shares with the estimator stays invisible to it, and the simulated coverage remains the figure a reader should weigh. We tested that explanation rather than leaving it as an argument. Running the same pre-pivoting inside the four simulated cells that carry the shortfall, with 1,000 replicates and 399 draws at each level, gives coverage of 0.924, 0.933, 0.931 and 0.944, against 0.938, 0.931, 0.936 and 0.953 for the shipped interval on the same replicates. An uncalibrated interval at the same 399 draws covers 0.936, 0.928, 0.938 and 0.949, so the draw count isn't doing the work. The calibration tracks the interval it corrects and repairs none of the shortfall, which is what a correction blind to the violation would do.

The estimator assumes that item noise in one half doesn't correlate with item noise in the other, so we asked how much a violation could manufacture. Holding the true seed correlation at zero, we swept the within-benchmark cross-half noise correlation from -0.1 to 0.5 with 1,000 replicates at each value. The mean margin estimate stays between 0.997 and 1.001 across that whole range, and no margin replicate reaches the observed 1.244. What the correlation changes is dispersion, which falls from 0.057 at zero to 0.046 at 0.5 on margins. Noise correlated this way adds the same quantity to the numerator and the denominator, so it can't create inflation and instead pulls a real inflation toward one, which is the -0.014 bias that Table 16 records. On accuracy the null dispersion is wide enough that 0.19 of replicates reach the observed 1.078 with no correlation at all, and that share falls to 0.07 at a correlation of 0.5 . This sweep says nothing about noise shared across benchmarks within a half and repeated in the opposite half, which adds the same quantity to every entry of the cross-half covariance, so the numerator absorbs all of it while the denominator absorbs only the diagonal tenth.

We measured that second violation in the same way as the first. Each half received a scalar effect that every benchmark loads on, repeated exactly in the opposite half, and we swept its share of item-noise variance with 2,000 replicates at each value and no seed covariance. The mean margin estimate climbs monotonically from 0.999 at a share of zero to 1.606 at a share of 0.4 , and a share near 0.124 reproduces the observed 1.244. At a share of 0.05 only 0.007 of replicates reach the observed value,

so small amounts of this contamination don't explain the margin result. Accuracy is far more fragile, because the same construction reaches the observed 1.078 at a share near 0.009, which is about 1 percent of item-noise variance. We haven't measured this component in the released scores, so these figures give the contamination a reader would have to assume rather than a bound we established. The passage-aware split addresses shared structure inside one benchmark and leaves this cross-benchmark channel open. We tried to detect the contamination in the data and failed. Regressing the off-diagonal entries of the estimated seed covariance on the item-noise outer product gives a slope of 1.675 on margins, with a recipe bootstrap interval of 0.534 to 3.068, and 0.033 on accuracy with an interval of -0.006 to 0.073 . Simulations with the observed covariance and no shared effect give a mean slope of 1.682 on margins, and adding the shared effect at the share that reproduces the observed estimate moves that mean only to 1.808, against a replicate spread of 1.290 to 2.371. The statistic can't separate the two explanations, because a seed covariance that follows the item-noise scales looks the same as contamination that follows them. We therefore report that diagnostic as having no power rather than as a clean check.

One part of the assumption does admit a direct test, and we ran it. Write the per-item score of run r in configuration c as x_{cri} , and let μ_{ci} be its mean over that configuration's three runs, so that $d_{cri} = x_{cri} - \mu_{ci}$ carries the seed effect together with the item noise. Permuting d across the items of one group, separately for every run, leaves each group mean and therefore each trait score exactly where it was, and it leaves any component that all items of a group share exactly where it was as well, because permuting a constant returns the constant. What the permutation destroys is the tie between particular items and particular runs, which is the channel a shared passage or a shared source document travels through. Over 2,000 permutations of the 37,682 items in their 66 groups, the margin estimate has mean 1.2428 with a standard deviation of 0.0009, and 0.087 of the permutations reach or exceed the observed 1.244, with a Monte Carlo standard error of 0.006 on that share. On accuracy the permutation mean is 1.0763 with a standard deviation of 0.0053, and 0.352 of them reach or exceed the observed value. The item-level channel therefore accounts for about 0.001 of the margin estimate and 0.002 of the accuracy estimate, against excesses over one of 0.244 and 0.078. Re-drawing the item half split 200 times moves the estimate further on the real data than under permutation, where the real split standard deviation of 0.0018 on margins beats the 0.0009 the permuted data give, and 0.0083 on accuracy beats 0.0055, so item-level structure does survive and it widens the spread without moving the centre. The shipped split returns 1.244 on margins against a re-split median of 1.246 and a range of 1.242 to 1.254, and 1.078 on accuracy against a median of 1.083 and a range of 1.059 to 1.103, so the reported numbers aren't a favourable draw. An effect that every item of a benchmark carries survives this permutation untouched, so the whole-benchmark form of the violation stays untested and only the sub-benchmark form is bounded here.

The coverage table uses a generic generator, so we also built two scenarios fitted to these data, each with 2,000 replicates and a Monte Carlo standard error near 0.005. Both draw a per-band seed covariance from the nearest positive semidefinite matrix to the observed estimate in each size band, and both simulate 125 configurations with shared item halves. The second scenario also scales each recipe's seed deviations by the square root of that recipe's observed denominator sum divided by the mean recipe denominator. The first scenario doesn't reproduce the observed concentration, because the observed margin influence share of 0.520 exceeds 0.990 of its replicates, while the second puts that share at its 60th percentile. In the first, margin coverage is 0.954 for the wild interval, 0.946 for the cluster-robust t , 0.954 for the centred inversion, and 0.970 for the restricted inversion, with 0.947, 0.940, 0.951, and 0.966 on accuracy. In the second, the wild interval holds 0.946 and 0.945 while the cluster-robust t falls to 0.901 and 0.927, and the centred inversion turns conservative at 0.977 and 0.958. We later reran both scenarios with two further constructions and a fresh seed, which reproduces every figure above inside two Monte Carlo standard errors and puts the wild interval at 0.946 and 0.942 on margins. The residual-scaled version keeps its advantage there, covering 0.960 in the first scenario and 0.964 in the second against those two, so the alternative that the generic generator preferred also wins under a description fitted to these data. Correcting the cluster-robust t by $G/(G-1)$ doesn't rescue it, because it reaches only 0.939 and 0.909 on margins, which leaves the gap against the wild interval where it was. These scenarios take their covariance, their item noise, and their recipe weights from the same data that the intervals describe, so they show how each procedure behaves under a fitted description of this population and they aren't independent confirmation.

On the observed data the centred inversion gives 1.095 to 1.323 for margins and 0.995 to 1.162 for accuracy, beside wild intervals of 1.143 to 1.338 and 0.993 to 1.157. At the full battery the upper endpoints differ by at most 0.015 and the two constructions agree on which intervals include one, but the inversion puts the margin lower endpoint 0.048 lower, so the wild lower endpoint shouldn't be read as a firm floor. In the size bands, the inversion gives 0.765 to 1.737 for 530M margins, which includes one although the wild interval excludes it, and 1.024 to 1.398 for 530M accuracy, which still excludes one. For 1B margins, where the wild lower endpoint is undefined, it gives a bounded set of 0.734 to 2.244. The accuracy intervals after BoolQ and WinoGrande removal stay above one under inversion, at 1.397 to 1.716 and 1.026 to 1.184.

Table 16. Coverage of nominal 95 percent intervals across 2,000 simulated populations per cell. Bias is the mean estimate minus the population Λ . Noise gives the item-noise standard deviation relative to unit run effects. Inversion is the centred cluster test-inversion interval, computed later on the same simulated moments.

Population	Λ	Bias	Wild	Cluster t	Config.	Recipe	Inversion
Null, noise 0.73	1.000	0.000	0.953	0.946	0.948	0.937	0.953
Null, noise 1.45	1.000	-0.011	0.943	0.941	0.943	0.857	0.946
$\bar{r}_E = 0.061$, noise 0.73	1.245	0.001	0.947	0.943	0.938	0.929	0.948
$\bar{r}_E = 0.018$, noise 1.45	1.078	-0.007	0.943	0.940	0.942	0.910	0.945
One trait with 0.68 of trace	1.152	-0.001	0.938	0.936	0.937	0.925	0.941
One trait with 0.84 of trace	1.031	0.000	0.954	0.950	0.955	0.940	0.957
Recipe-shared run effects 0.5	1.245	-0.003	0.944	0.940	0.872	0.931	0.947
Student t , 4 df	1.245	0.000	0.946	0.930	0.943	0.927	0.955
One recipe at 6 times variance	1.245	-0.001	0.930	0.911	0.931	0.912	0.955
Cross-half correlation 0.1, noise 0.73	1.245	-0.014	0.928	0.926	0.930	0.918	0.932
Cross-half correlation 0.1, noise 1.45	1.078	-0.016	0.946	0.941	0.943	0.933	0.948
$\bar{r}_E = 0.2$	1.673	-0.002	0.947	0.943	0.943	0.928	0.948

The analysis also reports 1,000 trait-specific run-label permutations, each using 4,999 bootstrap draws, whose intervals contain the reference value one in 0.925 of margin permutations and 0.954 of accuracy permutations, falling below one in 0.049 and above one in 0.026 against nominal tail probabilities of 0.025. We repeated that experiment for this revision with 2,000 permutations, which gives coverage of 0.944 on margins and 0.949 on accuracy with a Monte Carlo standard error near 0.005, and tail shares of 0.039 below and 0.018 above. That looked like a two standard error gap from the record, so we ran the experiment ten further times at the record's own 1,000 permutations, each on a separate stream. Margin coverage across those ten runs falls between 0.915 and 0.942 with a mean of 0.931, and accuracy falls between 0.925 and 0.959 with a mean of 0.944. All four recorded coverages, including the cluster-robust ones of 0.881 and 0.941, land inside the matching range, so the record reads as one draw at a replicate count too coarse to separate these values. The ten runs together put margin coverage 0.019 below nominal, which agrees with the 0.933 to 0.953 that the simulation gives. The lower-tail imbalance matters for interpreting an upper bound below the proposed threshold of 1.349, where the reported upper endpoint leaves only 0.011 of separation, approximately one quarter of a margin standard error.

These are conditional randomisation calibration results, contingent on the exchangeability assumed by the permutations. They aren't coverage measurements for the unknown nonzero covariance of the original population. The cluster- t comparator has recorded coverage of 0.881 and 0.941 with above-one rates of 0.044 and 0.023, and our 2,000-permutation repeat gives 0.901 and 0.938 with above-one rates of 0.034 and 0.023. In the record and in the repeat alike, no margin estimate reaches the observation, while the accuracy upper-tail fraction is 0.021 in the record and 0.020 in the repeat, close to 0.028 in the earlier 500 draws. With a plus-one correction, zero exceedances among 1,000 draws give a Monte Carlo value of 1/1001, rather than zero.

The reported delete-one-recipe jackknife standard errors are 0.049 and 0.039. Applying a two-sided 95 percent $t(24)$ critical value to those rounded inputs gives approximate intervals of 1.143 to 1.345 for margins and 0.998 to 1.158 for accuracy. The corrected accuracy interval includes one, consistent with the primary interval.

Across the 1,000 permutations, estimator standard deviations are 0.0488 for margins and 0.0382 for accuracy. The earlier 500-draw values are 0.048 and 0.039, compared with N1 values of 0.046 and 0.048. The accuracy discrepancy therefore concerns simulated versus conditional permutation populations, rather than establishing an error in the observed bootstrap scale.

Availability of a separate competence measurement The desired gain control requires per-run loss on held-out text separate from the benchmark items. The DataDecide fields inspected in the source analysis don't supply it. The PolyPythias evaluation files considered there cover three sizes at early and final checkpoints, but contain accuracies, likelihood differences, and LAMBADA perplexity on the same benchmark items. That perplexity doesn't provide the required independent measurement scale.

The Mistral release of Karamcheti et al. (2021) provides five GPT-2 Small and five GPT-2 Medium runs with about 600 checkpoints each, listed as 610 in the release summary, including WikiText-103 and OpenWebText validation perplexity. The source search didn't find the corresponding per-run benchmark battery. Without paired held-out loss and benchmark scores from the same runs, we can't use that release to identify the gain contribution.

Competence proxy audit and calibration We audited the shipped competence adjustment and two repairs on the 375 reduced runs, then applied each adjustment to simulated populations with known seed covariance. The audit reproduces the shipped after-adjustment estimates of 1.244 and 1.076. The variance floor binds for WinoGrande on margins and for PIQA and WinoGrande on accuracy, and WinoGrande then supplies more than 99.99 percent of the squared standardised inputs to the margin proxy, so the shipped margin adjustment in effect regresses every other benchmark on WinoGrande. Dropping each benchmark from the proxy in turn leaves margin inflation between 1.244 and 1.245, except for WinoGrande, whose removal gives 0.890 with interval 0.805 to 0.968.

The repair we fixed before seeing its output standardises by seed variances shrunk halfway toward their median positive value, estimated on four recipe folds and applied to the fifth. It gives margin inflation of 0.731 with interval 0.673 to 0.784, and accuracy inflation of 1.004 with interval 0.925 to 1.075. Cross-fitting without shrinkage leaves the floor binding and gives margin inflation of 1.245.

For this revision we fitted four proxy weightings out of fold, learning each variance, loading, and slope on four recipe folds and applying them to the fifth. A split-half check explains what the shipped proxy measures, since its half-A and half-B scores correlate at -0.086 on margins and -0.072 on accuracy, so the shipped proxy carries no measurable signal about the run it scores. The shrunk proxy reaches 0.913 on margins and 0.301 on accuracy, and a one-factor proxy with regression-score weights reaches 0.965 and 0.738. That factor puts 0.353 of its weight on one benchmark for margins and 0.467 for accuracy, and its fold weight vectors correlate at least 0.847 with each other.

Cross-fitted margin inflation is 1.245 under the shipped weighting, 0.731 after shrinkage, 0.973 for the factor proxy, and 0.889 when the clipped benchmarks are dropped, while the same four give 1.073, 1.004, 1.017, and 0.994 on accuracy. A recipe bootstrap of 999 draws, refitting the whole cross-fitted pipeline in each draw, gives 0.935 to 1.018 for the factor proxy on margins and 0.970 to 1.104 on accuracy, so the adjusted margin estimate stops separating from independence. Under simulated independent seed deviations all four variants return a mean between 0.998 and 1.003, so none of them manufactures covariance where none exists. None of this identifies mediation, because every one of these proxies averages the same seed deviations that it adjusts and so removes shared covariance by construction. Simulations settle what those cross-fitted numbers mean, since they supply the value each proxy returns when no competence variable exists. With the observed seed covariance and no competence variable at all, the four weightings return margin means of 1.097, 0.745, 0.980, and 1.029, and accuracy means of 1.109, 1.048, 1.066, and 1.072. The observed cross-fitted values sit inside the central 90 percent of those simulations for the shipped, shrunk, and factor weightings on both scales, so none of them carries evidence of a competence mechanism beyond covariance removal. The clipped-drop weighting is the one departure, since its observed 0.889 on margins falls below a simulated fifth percentile of 0.908 and its observed 0.994 on accuracy falls below a fifth percentile of 1.003. Under a single common correlation of 0.178 the factor weighting returns 0.852 on margins, which is further below one than the matched simulations give, so these proxies can overshoot when the covariance is equicorrelated.

Table 17 shows why none of these values measures competence. Under independent seed deviations every adjustment returns a mean between 0.999 and 1.002, so the adjustments neither create nor remove covariance where none exists. With a single common correlation of 0.178, which reproduces the observed margin inflation, the repaired margin adjustments return 0.875 to 0.948, below one. With the observed seed covariance they return 0.745 to 0.889, each within 0.015 of the observed 0.731 to 0.890. On accuracy the repaired adjustments lower the matched simulated mean from 1.116 to between 1.049 and 1.079, and the observed values of 1.004 to 1.019 lie inside the central 90 percent of those simulations. The repaired proxies average the same seed deviations that they adjust, so they remove shared covariance by construction and overshoot on margins, and the matched simulations reproduce their observed values from the covariance estimate alone. A competence adjustment therefore needs a measurement taken outside the benchmark scores, such as held-out loss on the same checkpoints.

We then checked whether those checkpoints are reachable. An index built from the model repositories matched 120 of the 375 runs to the training branch whose released step set overlaps the run’s evaluation steps, with a median overlap of 1.0 and a smallest overlap of 0.042, and each of those 120 runs has a released checkpoint at its selected step. The other 85 of the 125 repositories answer an unauthenticated request with HTTP 401, so we can’t download their weights and can’t score a held-out loss for the 255 runs they hold. Among the 120 matched runs, the branch’s final step disagrees with the run’s final evaluation step in 21 cases, which is a further obstacle to scoring loss at the step the scores describe. We report this as the reason the external measurement stays unavailable here, and not as evidence about what such a measurement would show.

Table 17. Inflation after each competence adjustment, observed and in simulated populations with 300 replicates each. Simulated columns give replicate means under independent seed deviations, a single common correlation whose population inflation equals each phenotype’s observed value, and the nearest positive semidefinite matrix to the observed seed covariance.

Adjustment	Margin				Accuracy			
	Observed	Indep.	Common	Matched	Observed	Indep.	Common	Matched
None	1.244	1.000	1.242	1.243	1.078	0.998	1.093	1.116
Shipped, floored	1.244	1.000	1.170	1.093	1.076	0.999	1.092	1.109
Floored traits dropped	0.890	1.001	0.948	0.889	1.008	0.999	1.083	1.079
Shrunk, in sample	0.732	1.002	0.876	0.747	1.019	0.999	1.061	1.050
Shrunk, cross-fitted	0.731	1.002	0.875	0.745	1.004	0.999	1.060	1.049

Transport to PolyPythias The planned transport analysis scores nine runs per configuration on a nested 4,755-item battery. For three sizes, the source pipeline gives margin inflation of 1.40574 compared with 1.24395 for DataDecide, an excess ratio of 1.66323 computed from unrounded values. A scoring-kernel rewrite reproduces both inflation estimates to five decimals and reports the ratio as 1.66321, a difference that arises beyond the fifth decimal of the inputs. The configuration table requires the second-arm output rather than inspection of the DataDecide reduction alone.

On excess inflation, the DataDecide prediction falls approximately 40 percent below PolyPythias, while the raw inflation difference is approximately twelve percent. The planning document names aggregate standard deviation without resolving this scale choice. The excess-based failure assessment is therefore post hoc, and the raw inflation comparison falls inside the proposed twenty-percent band.

A later exploratory extension includes the 14M and 31M configurations, each with nine runs at step 143000 on the same 4,755 items. Within that rescore, margin inflation changes from 1.41372 at three sizes to 1.26199 at five sizes, while the excess ratio changes from 1.69592 to 1.07393. Accuracy changes from 2.28589 to 1.71085, compared with DataDecide’s 1.07837. Only the three-size accuracy point estimate exceeds the stated detection threshold of approximately 2.05. The configuration-bootstrap interval for five-size accuracy runs from 0.869 to 2.428, and at three sizes its lower endpoint is undefined, with an upper endpoint of 2.971.

The margin configuration-bootstrap intervals are 0.939 to 1.781 at three sizes and 0.927 to 1.537 at five sizes. Both contain one and DataDecide’s 1.24395, while the standard error falls from 0.190

to 0.150. These intervals limit conclusions about transport under either point-estimate assessment. The configuration bootstrap treats three or five configurations as enough to resample, so we also inverted the centred cluster test of Appendix C with each configuration as its own cluster and $t(2)$ or $t(4)$ critical values. That construction gives margin sets of 0 to 2.106 at three sizes and 0.519 to 1.664 at five sizes, and it can't exclude any nonnegative accuracy value at either, so these data don't bound PolyPythias accuracy inflation at all. The rescore used one T4 with a 6,000-token cap instead of 30,000 tokens, and its three-size margin value of 1.41372 differs from the pipeline's 1.40574 by 0.6 percent. The excess-ratio difference is approximately two percent, with hardware and batching changed together. The three-versus-five comparison holds those conditions fixed within one job, and the source supplement describes 45 per-run score files from that job. For this revision we repeated the rescore on two T4 GPUs with the same scorer, token cap, items, and checkpoints. The repeat reproduces every estimate, configuration-bootstrap interval, and standard error that the earlier supplement reported, to the stated precision, and it corrects the five-size excess ratio, which the earlier supplement gave as 1.07395. We computed the inversion sets from the repeat's own score files, because the earlier supplement didn't report any inversion sets.

PolyPythias carries nine seeds for each configuration, which is enough to measure the ratio directly and to ask whether a three-run fit predicts it. The measurable quantity is the standard deviation of the battery mean over a set of runs, divided by the square root of the weighted sum of per-benchmark variances over the same runs, and it includes item noise. For each of the 84 ways of choosing three runs, we formed three predictions of that quantity on the six runs left out. Independence predicts one, the plug-in predicts the inflation estimate itself, and the decomposition adds each benchmark's estimated item noise to both sides of the ratio. On margins across three sizes the mean absolute log error is 0.339 for independence, 0.536 for the plug-in, and 0.512 for the decomposition, so neither fitted prediction beats the constant. On accuracy the same three errors are 0.370, 0.447, and 0.384, and the decomposition beats independence in 0.50 of splits. The five-size set repeats this ordering, and the decomposition beats the plug-in in 0.81 and 0.75 of margin splits, which is what the item-noise term is meant to do. The six-run target is itself noisy, because the measured margin ratio runs from 0.70 to 1.85 between its fifth and ninety-fifth percentiles, so this compares predictions of one noisy realisation rather than of the underlying ratio. Within that limit, the decomposition doesn't show out-of-sample value against a single configuration's ratio. Pooling the target across configurations changes the answer, which shows how much of the first comparison was target noise. When the six held-out runs give one ratio built from mean within-configuration variances, its standard deviation across splits falls to 0.140 on three-size margins. The decomposition then has mean absolute log error 0.275 against 0.319 for independence on three-size margins, and 0.367 against 0.523 on accuracy, and it wins in 0.60 and 0.73 of splits. The five-size set gives 0.209 against 0.218 on margins and 0.251 against 0.364 on accuracy. The decomposition also beats the plug-in on every set, by between 0.010 and 0.105 of log error, so carrying item noise into the prediction helps on all four sets, where the covariance term helps on three of them and hurts on five-size margins. The 84 splits share runs and the four comparisons share configurations, so these are descriptive differences without an interval.

Table 18. Five-fold prediction error for log aggregate seed standard deviation. Every model receives held-out trait standard deviations, so the comparison measures correlation prediction with variance estimation fixed.

Model	Parameters	Margin error	Accuracy error
P0 independence	K	0.0358	0.0143
P1 rank-one competence	$K + 1$	0.0057	0.0140
P1 G rank-one plus gain	$2K + 2$	0.0063	0.0260
P2 performance correlation	K	0.1073	0.1682
P3 full covariance	$K(K + 1)/2$	0.0068	0.0105

Table 18 gives fold-mean point errors, and Appendix C gives paired intervals for their differences. P1 and P1 G have lower point errors than P3 on margins and higher errors on accuracy, while P0 and P2 have higher errors on both scales. P2 uses performance correlation from the held-out fold itself because its calculation needs no seed replication. Negative held-out variance estimates are clipped to zero before model selection, which also applies to independence. On the complete population this

clips one margin and two accuracy traits, increasing the aggregate covariance sum by 0.3 and 4.8 percent, while the scored target retains the original estimates. Shared clipping doesn't guarantee that it leaves model rankings unchanged, so we repeated the comparison under two alternative rules, with the same 2,000-draw paired bootstrap as Appendix C. On margins, the variance is nonpositive for OpenBookQA and WinoGrande in one fold and for WinoGrande in two more. Dropping an affected trait from that fold's prediction, correlation, and target gives margin errors of 0.0063 for both P1 and P3, and filling its held-out variance with the training-fold estimate gives 0.0039 and 0.0045, while independence stays near 0.035. Under both rules the P1 G and P3 intervals for a difference from independence still exclude zero, but under training fill the P1 interval runs from -0.059 to 7×10^{-5} and so includes zero. On accuracy, one to three traits are affected in every fold, and dropping them leaves P3 the best model, with its error falling from 0.0105 to 0.0090, while filling gives P1 and P3 errors of 0.0122 and 0.0118. None of the accuracy intervals for these contrasts excludes zero under either rule.

External random-seed panel The signal-and-noise runs don't share models with the DataDecide analysis, although the releases come from one laboratory, use a shared scoring harness, and overlap in all ten study traits. The check averages estimates over twenty released checkpoints and uses ten initialisation runs or nine data-order runs. It supplies single-unit estimates without cross-half correction or uncertainty intervals, which limits comparison with the corrected primary estimator.

Table 19. Inflation estimates in the external random-seed panel across twenty checkpoints. The leave-one-out adjustment controls each trait deviation with the mean deviation of the other traits.

Seed variation	Phenotype	K	Λ	Adjusted Λ	$\bar{r}_E$
Initialisation	bits per byte	18	1.705	1.464	0.112
Initialisation	accuracy per character	8	0.989	1.105	-0.003
Data order	bits per byte	18	1.625	1.763	0.096
Data order	accuracy per character	8	0.868	1.059	-0.035

The likelihood estimates exceed one under both seed mechanisms, while the unadjusted accuracy values lie near or below one. Adjustment increases three of the four estimates and reduces only initialisation likelihood, from 1.705 to 1.464. That pattern doesn't support a stable interpretation of the adjustment at one unit. Independent item noise would attenuate covariance ratios toward one under the relevant assumptions, but this arm doesn't measure the amount of attenuation or verify those assumptions. We therefore use this panel to compare score scales, with the size of item-noise attenuation still unmeasured.

C.1 GAIN CALIBRATION AND RECIPE SPLIT SENSITIVITY

The default gain simulator applies a shared margin scale across traits. The matched simulator instead copies each benchmark's mean margin, item-noise variance, and seed variance from the data, and it keeps seed deviations independent across benchmarks. At $\sigma_g = 0.05$, the two reported excess shares differ by a factor of about 7.5. Subtracting estimated leakage variance 1.91×10^{-3} from total gain variance 3.67×10^{-3} gives a gain standard deviation of approximately 0.042. This subtraction depends on the variance decomposition assumed for the covariate.

Table 20. Gain-simulator sensitivity relative to the observed margin excess of 0.244. These shares compare specified simulations and don't bound a causal contribution in the observed runs.

σ_g	Default $\hat{\Lambda}_M$	Excess share	Matched $\hat{\Lambda}_M$	Excess share
0.01	1.007	0.029	1.001	0.003
0.02	1.026	0.108	1.003	0.014
0.03	1.058	0.236	1.008	0.031
0.042, estimated	1.108	0.443	1.014	0.059
0.05, default	1.149	0.609	1.020	0.081
0.10	1.465	1.905	1.062	0.256

The default 0.05 cell differs by approximately 0.0009 from the corresponding Table 12 result before rounding, less than half the reported Monte Carlo standard error for 500 replicates. The G5 assessment uses that default specification, which fails its threshold. Accuracy estimates stay within 0.004 of one in these simulations, but positive multiplication preserves signs by construction. Their proximity to one therefore doesn't establish absence of a gain mechanism for observed accuracy variation.

To carry uncertainty in the gain scale into these shares, we resampled the 25 recipes 999 times and re-estimated the gain standard deviation in each draw. The median is 0.041, the 95 percent percentile interval runs from 0.011 to 0.063, and the leakage-corrected variance is nonpositive in 0.9 percent of draws. With 300 replicates at each endpoint, the default simulator gives margin inflation of 1.006 and 1.221, which are shares of 0.02 and 0.91 of the observed excess, while the matched simulator gives 0.999 and 1.028, with shares of -0.01 and 0.12. At the point estimate, 300 new replicates give 1.105 and 1.012, within two Monte Carlo standard errors of the 500-replicate values in Table 20.

We also compared each simulator's populations with the observed runs at the point estimate. The default simulator works on a standardised scale, with seed variances between 1.02 and 1.05 and mean margins near 3.85, whereas the observed seed variances lie below 0.002 and the observed mean margins lie between -0.18 and 0.07, so its shares depend on a ratio of margin level to seed noise that it assumes rather than measures. The matched simulator reproduces the observed mean margins and item-noise variances of every benchmark to within 1.2 percent, and the seed variances of the four benchmarks with the largest estimates to within 6.1 percent. Its margin configuration means vary far less than the observed ones, with standard deviations between 0.02 and 0.45 times the observed values, because it adds no systematic differences between configurations. Its accuracy seed variances exceed the observed estimates by factors of 2.4 to 116 on the eight benchmarks where those estimates are positive. When we apply the gain estimator to the simulated populations, it recovers a mean of 0.037 under the default simulator and 0.036 under the matched simulator, 11 and 15 percent below the input of 0.042, so the estimator itself understates the gain scale in both specifications.

For a recipe subset S , the estimator is $\hat{\Lambda}_S = \sqrt{\sum_{c \in S} T_c / \sum_{c \in S} U_c}$, because each configuration's components require no other configurations. The analysis enumerates all $\binom{25}{17} = 1,081,575$ possible seventeen-recipe estimation sets. It reports completing this sweep in thirteen seconds on one CPU core. The reconstructed split uses the plan's master seed under the released implementation, but the plan didn't specify a complete reconstruction rule and no contemporaneous split artifact was recovered.

Table 21. Exhaustive sensitivity to the choice of seventeen estimation recipes. Every admissible subset gives a finite point estimate under both phenotypes.

Quantity	Margin	Accuracy
All 25 recipes	1.244	1.078
Reconstructed planned partition	1.199	1.099
Minimum	1.037	0.963
5th percentile	1.176	1.035
Median	1.246	1.078
95th percentile	1.285	1.124
Maximum	1.317	1.191
Effective-count range	[5.77, 9.31]	[7.05, 10.77]
Planned threshold	1.349	1.40
Subsets reaching threshold	0	0
Subsets below one	0	1,393, or 0.13%

No subset reaches either proposed point threshold in this enumeration. The sweep measures sensitivity to recipe selection after the analysis choices were made, so it provides no check on choices that used the full dataset.

That enumeration moves the point estimate, and it doesn't say whether the reported conclusions themselves survive a deletion, so we recomputed the shipped interval after removing each recipe,

each size band, and each benchmark in turn. For the full-battery margin we also removed all 300 pairs of recipes. The margin interval keeps excluding one under every one of the 25 single-recipe deletions, under all 300 pairs, and under every single-benchmark deletion, where the estimate over the pairs stays between 1.177 and 1.263 and the nearest endpoint sits 0.037 above one. One deletion breaks it, because dropping the 530M band leaves 1.178 with an interval of 0.993 to 1.338, so the lower endpoint falls 0.007 below one on the remaining 100 configurations. We calibrated that exception the same way we calibrate the accuracy side. Simulating 10,000 margin-like populations over two runs with separate seeds, and keeping the 9,697 whose full interval excludes one, at least one of the five band deletions loses the exclusion in 0.164 of them, and at least one of the 25 recipe deletions loses it in 0.041. The two runs agree on the band figure inside a quarter of a standard error of their difference and on the recipe figure inside one and a quarter, and the later run carried 8,000 of those replicates. One band failure out of five is therefore close to what a deletion sweep of this design produces by chance, which is a weaker statement than the deletion looks and a fairer one. The same simulator answers the question a reader planning their own replicate study would ask, which is how much data an effect of this size needs. At our 125 configurations and three runs the margin effect clears one in 0.970 of replicates while the accuracy effect clears it in 0.111, so the two conclusions differ in power before they differ in substance. Holding three runs and raising the recipe count to 50, 100 and 200 takes the accuracy figure to 0.165, 0.306 and 0.620. Six runs per configuration take the same four designs to 0.196, 0.398, 0.704 and 0.947. Four fifths therefore needs roughly 500 configurations at six runs, which is eight times the run budget of this release, and that number is the honest answer to why our accuracy interval includes one. One deletion remains that none of this covers, which is the deletion of a run. Each configuration holds three of them, so a single anomalous run has less data standing between it and the estimate than any column of the design. Re-estimating on each surviving pair leaves the margin estimate at 1.256, 1.242 and 1.235, all with intervals that exclude one, and the accuracy estimate at 1.084, 1.114 and 1.030, none of which excludes it. Two runs leave one contrast direction instead of two, and the margin interval widens from 0.195 to between 0.201 and 0.356, which is the cost of that lost direction rather than a change in the answer. The accuracy conclusion is the fragile one, and it is fragile in the opposite direction. Its full-battery interval includes one, and it stops including one under five of the 25 recipe deletions, under the removal of the 750M band, and under the removal of WinoGrande as well as BoolQ. All eight of those exceptions put the whole interval above one rather than below it, seven of them with lower endpoints between 1.002 and 1.025, and the eighth is the BoolQ removal that we already report at 1.412. Twenty-five correlated deletions of the same data give plenty of chances to flip something, so we calibrated the count against a null rather than reading it directly. Simulating 10,000 populations whose true ratio is exactly one under accuracy-like noise, again over two runs with separate seeds, and keeping the 9,477 whose full interval includes one, at least one recipe deletion flips in 0.105 of them and five or more flip in 0.018. A second null calibrated to the observed 1.078 puts those at 0.127 and 0.032 over its 8,964 kept replicates. All six of these calibration figures reproduce across the two runs inside 1.8 standard errors of their difference, so the larger repeat sharpened them without changing what they say. The observed five recipe flips therefore sit in the upper few percent of what a deletion sweep produces by chance, which is why we read them as a direction rather than as noise. The deletions that move the accuracy conclusion move it towards more inflation, so a reader should treat the inclusion of one on the full battery as a statement about that battery rather than about accuracy in general.

Because one size band carries that much weight, we also asked whether clustering on recipes alone leaves the interval too narrow. Clustering the same linearised residual on the five size bands instead gives a standard error of 0.175 on the squared margin ratio against 0.112 from the recipes, and its interval runs from 1.030 to 1.426 with four degrees of freedom and a critical value of 2.776, so it still excludes one. The matching accuracy interval runs from 0.961 to 1.184 and still includes one, which means both full-battery conclusions survive a clustering on the dimension the deletions flagged. The two-way variance of Cameron, Gelbach and Miller doesn't help here, because the component for the intersection of recipe and size band, which puts each configuration in its own cluster, gives 0.233 and exceeds the sum of the other two, so the two-way variance turns negative on the full-battery margin and no interval exists. On full-battery accuracy the same subtraction returns a standard error under half the recipe one, which we don't believe and don't report as a result. The estimator behaves on the battery without BoolQ, where it widens the recipe standard error by factors of 1.29 and 1.40 and leaves both scales above one, at 1.652 to 1.910 and 1.276 to 1.796. A bootstrap version of the same correction has no negative-variance failure mode, because it multiplies each residual by the

product of one Rademacher draw per dimension and reads the interval off the resulting spread, so we ran it at 9,999 draws. It returns 1.051 to 1.411 on the full-battery margin, which is the cell the analytic version couldn't deliver, and it still excludes one. On full-battery accuracy it returns 0.973 to 1.175, which is wider than the recipe interval rather than narrower, so the analytic accuracy figure we declined to report was the artefact we took it for. The bootstrap doesn't repair the failure mode itself. We first measured that over 1,000 replicates at 999 draws, which left both comparisons inside their own Monte Carlo noise, so we repeated the two cells at 8,000 replicates and 2,999 draws and scored the two intervals on the same replicate each time. At a band-shared quarter the two-way bootstrap covers 0.823 against 0.833 for the recipe-clustered interval, and at the joint corner the data allows it covers 0.932 against 0.940. Because the two intervals see identical data, the paired difference carries the comparison, and it comes to -0.010 with a standard error of 0.0024 at the quarter and -0.008 with a standard error of 0.0019 at the corner. Both sit more than four standard errors below zero, so the two-way interval is measurably worse here rather than merely no better, and the repeat also reproduces the four observed cells at 9,999 draws to every digit we report. We therefore keep the recipe clustering, and we note that five size clusters are too few for the analytic two-way correction at this design.

The released seed labels make the failure mode of that choice concrete, because the release uses seeds 2, 14 and 15 below 1B and seeds 2, 4 and 5 at 1B, so one seed label is shared across every recipe inside a size band. We checked that those labels mean what we take them to mean, rather than assuming it. The model repositories name their branches by step and by one of default, small-aux-2 and small-aux-3, and nothing in the release states which released seed each branch produced. Rescoring the first 150 ARC-Easy documents at 150M from the request file the release itself ships, and forming the same per-byte margins, matches the default branch to released seed 2, small-aux-2 to seed 14 and small-aux-3 to seed 15. It does so in all eight recipes whose repositories resolve, which are c4, falcon, fineweb-edu, fineweb-pro, dclm-baseline and its three blends with dolma 1.7, giving 24 branch cells without a single exception. Twenty of those matches carry a correlation of 0.99999997 or better with a median absolute difference between $1.0e-05$ and $2.0e-05$, the four weakest still reach 0.99993, and the best wrong seed in any cell is 0.897, so no cell is ambiguous. A separate run at 1B now covers five recipes rather than one. The 1B band is where the release changes both halves of the naming, shipping seeds 2, 4 and 5 and branches called large-aux-2 and large-aux-3, so it is the band where a silent difference would have hurt most. Rescoring the same 150 documents there matches the default branch to seed 2, large-aux-2 to seed 4 and large-aux-3 to seed 5 in all fifteen cells whose repositories resolve, without an exception. The weakest of those fifteen correlations is 0.99986 and the best wrong seed in any cell is 0.910, so no cell is close to ambiguous. The scores we analyse therefore reproduce from public weights, and the seed labels that this clustering argument leans on are the labels we believe they are. The other seventeen recipes never loaded, and a direct census of all 125 repositories separates the two reasons a load can fail. Eighty of them answer an unauthenticated request with HTTP 401, five carry the name dolma1.6 with two plus signs, which the hub rejects as an identifier, and 40 resolve. That split is identical in each of the five size bands, at sixteen refusals, one rejected name and eight open repositories. An anonymous client cannot tell a private repository from a missing one, so the loader reports both as an invalid identifier, and we record the status code rather than the message. The census also reads the branch list of every open repository, which carries default, small-aux-2 and small-aux-3 in all 40 and adds large-aux-2 and large-aux-3 in the eight at 1B, so the branch naming holds even where the weights stay closed. We simulated a run effect that follows the size band rather than the recipe, built exactly as the existing recipe-shared effect is built so that the true ratio doesn't move, and swept its share of latent variance with 2,000 replicates at each value, then repeated the three shares the limitation rests on at 10,000 more with a second seed. At a share of zero the recipe-clustered wild interval covers 0.943 and the size-clustered one covers 0.951 over the pooled 12,000, so the two agree when the assumption holds. At a quarter the recipe-clustered interval falls to 0.821, at a half it reaches 0.580, and at three quarters, which only the first pass ran, it reaches 0.348, while its median width shrinks from 0.242 to 0.189 because the recipe clusters never see the added variance. The size-clustered cluster-robust t holds 0.935, 0.898 and 0.758 over the same sweep. The two passes agree to within 0.013 at every cell, and the pooled figures are the ones we quote. A reader who thinks this dependence is present should therefore read the size-clustered interval on the observed data, which runs from 1.030 to 1.426 on margins and still excludes one, and from 0.961 to 1.184 on accuracy and still includes one. Both reported conclusions survive the clustering that is robust to this violation, and the reported interval's width doesn't. Recommending

that fallback obliges us to test it against the sharing channel these data actually bound, which is the recipe one rather than the band one, so we swept a recipe-shared run effect over 0, 0.114, 0.25 and 0.50 at 4,000 replicates with seed 20260955. The size-clustered wild interval covers 0.953, 0.955, 0.944 and 0.898 across that sweep, and its cluster-robust counterpart covers 0.947, 0.944, 0.929 and 0.869. At the 0.114 bound the permutation test puts on recipe sharing, the fallback therefore costs nothing at all, and it only starts to slip at a half, which sits more than four times that bound. What it costs instead is width, since five clusters leave a median width near 0.357 against 0.246 for the recipe-clustered interval on the same replicates. The recipe-clustered interval itself holds between 0.942 and 0.947 throughout, which is what a sweep of the dependence it already models should show, and the reimplemented interval reproduces the shipped one exactly in every cell. Two sweeps ask whether any of this survives a design that isn't the one the release happens to ship. Widening the battery from four benchmarks to twenty raises the true ratio from 1.088 to 1.469, because flat weights over a compound-symmetric truth put the squared ratio at one plus k minus one times the correlation, and the recipe-clustered interval covers between 0.941 and 0.951 at every width with no undefined replicate anywhere. Varying the runs per configuration from two to ten leaves that interval between 0.942 and 0.952, while the median width falls from 0.343 at two runs to 0.242 at the three the release ships and 0.114 at ten. The size-clustered fallback is the one that suffers at two runs, dropping to 0.915, so a reader with only two runs per model should keep the recipe clustering. Both sweeps ran at 4,000 replicates with seeds 20260956 and 20260957. A companion sweep varies the number of size bands rather than the runs, holding the cell count near 125 by pairing 41 recipes with three bands, 25 with five, 16 with eight and 10 with twelve. The recipe-clustered interval covers between 0.938 and 0.948 across all four, so the band count doesn't reach the clustering the paper reports. The fallback is the arrangement that cares, since the size-clustered wild interval covers only 0.900 with three bands and recovers to between 0.943 and 0.947 at five, eight and twelve, which says the fallback needs at least the five bands this release ships. A third sweep moves the item noise itself over 0.25, 0.50, 0.73, 1.00, 1.45, 2.00 and 3.00, which brackets both scales the paper reports and runs past them. The recipe-clustered interval covers between 0.944 and 0.955 from the quietest cell through 2.00, and it collapses at 3.00 to 0.668 with 0.312 of replicates undefined, so the estimator breaks somewhere near four times the margin battery's noise rather than anywhere near the accuracy battery's 1.45. Power is what the noise actually buys, because the exclusion rate at a true 1.2446 falls from 0.998 at the quietest cell to 0.630 at 1.45 and 0.326 at 2.00. These two ran at 4,000 replicates with seeds 20260977 and 20260976. A third sweep asks how many recipes the clustering itself needs, because 25 clusters is a small number for cluster-robust inference and the wild bootstrap is the standard answer to exactly that problem. Sweeping the recipe count over 5, 10, 15, 25, 40 and 60, the wild interval covers 0.749, 0.946, 0.942, 0.942, 0.946 and 0.946, so the shipped design sits well above the failure point rather than on it. The failure at five clusters is worth naming, since there the wild interval does worse than the plain cluster-robust one, which holds 0.910, and the usual advice would have predicted the opposite. What the extra clusters buy is power rather than coverage, because the exclusion rate at a true ratio of 1.2446 climbs from 0.54 at ten recipes to 0.97 at the shipped 25 and 1.00 at sixty. Two further sweeps break the assumptions the generator has been handing the estimator all along. Drawing the latent run effects from a multivariate t leaves coverage at 0.944 with ten degrees of freedom and 0.940 with five, and only a t with three, which has no finite kurtosis, pulls it down to 0.895. Giving the 25 recipes variance scales spread over a four-fold and a sixteen-fold range leaves coverage at 0.938 and 0.946, so unequal recipes cost nothing measurable. The last sweep prices the assumption the whole estimator rests on, which is that the two item halves carry errors that don't correlate. A correlation of 0.02 or 0.05 between them leaves coverage at 0.946 and 0.943, a correlation of 0.10 costs about a point at 0.928, 0.20 costs two at 0.920, and 0.40 costs seven and a half at 0.864. Our passage-level check moved the estimates by less than 0.001, which places the real leak far below the first of those steps, and the sweep says what a larger one would have cost. All three ran at 4,000 replicates with seeds 20260960, 20260961 and 20260962. Two more sweeps finish the picture of what this design can stand. Giving each benchmark the item noise its size implies, from the item counts the release ships, which spread from 500 on OpenBookQA to 14,042 on MMLU, leaves coverage at 0.945 against 0.946 for equal noise, and doubling that spread leaves it at 0.948, so the equal-noise simplification cost nothing. Crossing three bootstrap weight families with the cluster count says the same about the weight choice, since at the shipped 25 recipes Rademacher, Mammen and Webb weights cover 0.948, 0.939 and 0.948, and they only separate at five clusters where all three have already failed. Mammen weights do best in that failed corner at 0.873 against 0.830 for Webb and 0.760 for Rademacher, which is a curiosity rather than a recommendation, because no reading of a 0.873 interval is safe. The last question a

reader will have is what the within-size evidence would have needed, and the answer is runs rather than cleverness. A single band of 25 configurations at the shipped three runs covers 0.930 while excluding one in only 0.313 of replicates at a true ratio of 1.2446, which is why the 1B interval includes one, and six runs lift that to 0.739 and ten runs to 0.957 with the median width falling from 0.543 to 0.253. Ten runs per configuration inside one band would therefore have settled the question the paper has to leave open. These ran at 4,000 replicates with seeds 20260966, 20260967 and 20260963. Every sweep above hands the estimator a truth where all ten benchmarks share one correlation, and this battery is not built that way, so the last run measures the real covariance and sweeps against that instead. Ten single-benchmark restrictions give the diagonal and forty five pairs give the off-diagonal, all of it read through the shipped estimator rather than by hand, and the matrix that comes back reproduces the reported ratios to four decimals, at 1.2439 on margins and 1.0784 on accuracy. Projecting it onto the positive semidefinite cone clips one small negative eigenvalue on margins and three on accuracy, moving the ratios to 1.2438 and 1.0778. Simulating against those measured matrices leaves the recipe-clustered interval covering 0.947 on margins and 0.945 on accuracy, so the flat truth the other sweeps use was a harmless simplification rather than a flattering one. That run also prices the accuracy conclusion, because at a true ratio of 1.0778 the interval excludes one in 0.103 of replicates. A separate sweep of the true ratio at the accuracy battery's noise says the same thing in more detail, since the exclusion rate runs 0.018 at a ratio of exactly one, 0.063 at 1.05, 0.135 at 1.10, 0.281 at 1.15, 0.668 at 1.25 and 0.968 at 1.40. The accuracy battery could therefore only have found dependence reliably somewhere near 1.35, and a reader should treat its interval as evidence against a large ratio rather than as evidence for independence. Both ran at 4,000 replicates with seeds 20260965 and 20260964. The margin battery is the more sensitive of the two by a factor near two and a half, since at its own item noise the exclusion rate runs 0.014 at a ratio of one, 0.104 at 1.05, 0.332 at 1.10, 0.652 at 1.15 and 0.970 at 1.25. The point estimate itself is close to unbiased across that whole range, which matters because a ratio of sums under a square root has two reasons to run low. Over the same sweep the mean estimate sits between 0.0005 above and 0.0028 below the truth on margins, and between 0.0031 and 0.0067 below it on accuracy, against estimate standard deviations of 0.055 and 0.098. The bias is therefore under a tenth of a standard deviation everywhere, and no replicate in either sweep produced an undefined estimate. The bias sweep ran at 4,000 replicates per cell under seed 20260973. A last pair of runs tests the two things those sweeps still take on trust. Five explicit covariance shapes, a flat one, one giving a single benchmark 68 percent of the trace, one adding negative links from that benchmark to the other nine, one splitting the battery into two correlated blocks, and one combining dominance with negative links, leave coverage between 0.936 and 0.946. Two of those shapes carry a true ratio below one, at 0.9927 and 0.9900, and the interval excludes one in 0.011 and 0.009 of replicates there, so the estimator does not manufacture dependence when the truth runs the other way. The permutation null earns the word exact under the same abuse, since holding the true band share at zero and drawing the latent effects from a t with five and with three degrees of freedom, and spreading the recipe variances sixteen-fold, leaves the rejection rate between 0.034 and 0.058 against a nominal one in twenty. Those two runs used seeds 20260968 and 20260969 at 4,000 replicates each. We then asked the data how much of that effect it actually holds. Lining each configuration's runs up by their batch slot and averaging the run deviations over the 25 recipes of a band leaves a band-shared component untouched while shrinking anything specific to a recipe, so the weighted cross-half covariance of those band means measures the component directly. Permuting each configuration's runs among its own batch slots gives the exact null, because it preserves every configuration's own deviations and destroys only the alignment across recipes. Over 2,000 permutations the observed statistic sits inside the null on all four batteries, with 0.28 and 0.74 of permutations at or above it on the full battery and 0.13 and 0.36 without BoolQ. The permutation mean matches the independence prediction to within 1.3 percent everywhere, which lets us read the statistic as a share of the seed covariance. That share comes to 0.008 on full-battery margins with a 95 percent upper bound of 0.044, and to 0.020 without BoolQ with an upper bound of 0.053. A bound is only worth reading if the test behind it can see the thing it bounds, so we measured that on simulated populations whose band share we set, at 500 replicates and 999 permutations each. At a true share of zero the test rejects 0.044 of the time at one in twenty and 0.014 at one in a hundred, so it carries the size it advertises, and the plug-in share estimate misses the truth by 0.0009 or less at every share up to a quarter. Its power climbs from 0.218 at a share of 0.02 to 0.490 at 0.044, then 0.838 at 0.10 and 0.992 at a quarter, so the test would almost always catch the share that costs twelve points of coverage, while a share sitting at the bound itself would escape it about half the time. The standard error we quote is the standard deviation of the permutation null, and that understates the spread of the estimate once the true share is large, at

0.0216 against an actual 0.0386 at a share of 0.044 and 0.0219 against 0.0942 at a quarter. The bound is therefore trustworthy near zero, where the observed estimate of 0.008 sits and where the interval around it covers 0.962, and it runs anti-conservative out at the larger shares it is meant to exclude, so we read it as a limit on what these data can resolve rather than as a confidence statement about the truth. Because the smallest share in that first sweep was 0.25, we repeated the sweep at 0.02, 0.05 and 0.10 with 4,000 replicates each, which brackets the estimate and its bound and reaches twice the bound. The recipe-clustered wild interval covers 0.946, 0.941 and 0.929 across those three, with a Monte Carlo standard error near 0.004, so at the bound itself the interval loses about half a point of coverage rather than the twelve points it loses at a quarter. The mirror test, which averages over the five sizes of a recipe instead, also puts the observed statistic inside its null on all four batteries, with tail shares from 0.086 to 0.556. That version is far weaker, because five configurations per recipe leave a standard error of 0.067 on the share against 0.018 for the band version, so it bounds recipe-level sharing only below 0.114 on full-battery margins. Weaker does not mean blind, and we measured its power the same way we measured the band version's, sweeping the recipe share over 0, 0.05, 0.114, 0.25 and 0.50 at 500 replicates and 999 permutations each. The mirror test rejects 0.054 of the time when the true share is zero, its plug-in estimate misses the truth by 0.0046 or less at every share, and its power runs 0.248 at 0.05, 0.628 at its own 0.114 bound and 0.982 at a quarter. The bound the corner simulation leans on is therefore a measurement rather than an artefact, since a recipe share of a quarter would be caught in 98 replicates out of 100. Neither test speaks to the corner where both channels sit at their bounds together, so we simulated that corner as well, at 4,000 replicates per cell. Recipe sharing alone at 0.114 leaves the recipe-clustered interval at 0.949, and twice both bounds takes it to 0.927. The two cells that carry the limitation are band sharing alone at its 0.044 bound and the joint corner where both channels sit at their bounds together, and the first pass separated those by 0.012, at 0.937 against 0.949, which is a gap 4,000 replicates cannot resolve. We repeated both at 12,000 replicates with a fresh seed, and the separation doesn't survive. The larger run puts band sharing alone at 0.9444 and the joint corner at 0.9428, half a Monte Carlo standard error apart, with median widths of 0.242 and 0.246. Pooling the two passes over 16,000 replicates gives 0.9425 and 0.9444. Both corners therefore cost the interval between half a point and a point of coverage, and the earlier reading that the two sharing channels partly cancel rested on a difference the first pass was too small to measure. A remedy recovers even that small loss, and it costs almost nothing. Scaling each recipe's residual by one over one minus its leverage, which is the cr3 correction, lifts the recipe-clustered interval from 0.942 to 0.950 at the band corner and from 0.946 to 0.954 at the joint corner over 12,000 replicates, and it widens the median interval only from 0.241 to 0.252. We report the uncorrected interval in the main text because that is what the shipped estimator computes, and a reader who wants the corner protected can buy it for four percent of width. The sharing our clustering assumes is therefore a precaution that these data can neither confirm nor rule out, while the sharing that would break that clustering is bounded where it costs under a point of coverage.

D IMPLEMENTATION DETAILS

D.1 DATA STREAMING

The implementation streams 122.9 GB of gzipped JSONL tarballs from the Hugging Face release. It retains the item identifier, correct-answer margin, and accuracy bit for the ten traits at the common checkpoint. Each run contains 37,682 items and approximately 30 MB of compressed output. Reduction of 375 runs produces approximately 48 MB of NumPy archives, with one array per run.

A frozen master seed generates a balanced permutation of item order within each trait. The split uses that permutation rather than a hash of item identifiers, and all runs within an arm reuse the same trait-specific assignment. The PolyPythias subsample therefore has its own split, rather than an item-hash split shared across arms. Regenerating the split requires the recorded item ordering and randomisation specification, without downloading the full tarballs.

D.2 INTERVAL CALCULATION

Derived quantities throughout the paper are computed from unrounded estimates, so recomputation from the printed three-decimal values can differ in the final digit.

We state the source calculation in terms of the squared ratio,

$$\hat{\theta} = \frac{\sum_c T_c}{\sum_c U_c}, \quad \hat{\Lambda} = \sqrt{\hat{\theta}}.$$

For each recipe g among $G = 25$, the linearised contribution and standard error are

$$\psi_g = \frac{\sum_{c \in g} (T_c - \hat{\theta} U_c)}{\sum_c U_c}, \quad \widehat{\text{SE}} = \left(\sum_{g=1}^G \psi_g^2 \right)^{1/2}. \quad (20)$$

This standard error belongs to the squared-ratio scale used for studentisation. A reported standard error on the Λ scale instead requires the corresponding delta-method conversion.

For draw b , independent recipe signs satisfy $v_{bg} \in \{-1, +1\}$ with equal probabilities. Every configuration takes the sign assigned to its recipe, and the pseudo-numerator becomes

$$T_c^* = \hat{\theta} U_c + v_{bg(c)} (T_c - \hat{\theta} U_c).$$

The denominator remains fixed at the observed $\sum_c U_c$. We recompute $\hat{\theta}_b^* = \sum_c T_c^* / \sum_c U_c$ and the bootstrap standard error from the residuals $T_c^* - \hat{\theta}_b^* U_c$. The bootstrap statistic is $t_b^* = (\hat{\theta}_b^* - \hat{\theta}) / \widehat{\text{SE}}_b^*$, and the source calculation uses 4,999 draws with bootstrap seed zero.

Writing q_α for the empirical quantile of these statistics, the interval on θ is

$$\left[\hat{\theta} - q_{0.975} \widehat{\text{SE}}, \hat{\theta} - q_{0.025} \widehat{\text{SE}} \right].$$

Square-root transformation gives endpoints on the inflation scale when both endpoints are nonnegative. Monotonicity preserves the ordering of available endpoints, but it doesn't establish the original interval's coverage.

A draw with zero bootstrap standard error has no finite studentised statistic and is discarded. The routine requires at least $\max(100, 0.9B)$ surviving draws before reporting an interval. A negative lower squared-ratio endpoint is recorded as missing rather than truncated, which explains the absent 1B margin lower endpoint in Table 9. The configuration percentile bootstrap instead resamples all 125 configurations with replacement and recomputes both sums, while ignoring recipe membership.

The planning document specified wild cluster bootstrap-t inference with $t(16)$ critical values for seventeen estimation recipes. The realised analysis uses empirical bootstrap quantiles over all 25 recipes, following the failure to implement the planned split. The analysis record states that a $t(G - 1)$ variant accompanies the bootstrap implementation. These alternatives differ in more than a change of cluster count.

D.3 PRACTICAL CONSIDERATIONS

Within-configuration centring removes the shared performance mean before forming cross-half products. With only three runs, raw products would include a large contribution from mean competence rather than measuring covariance of deviations. The contrast representation in Appendix D.7 enforces that centring at the estimator input.

The permutation procedure reassigns run labels independently within each trait. Applying one joint permutation to all traits preserves cross-trait pairing and therefore doesn't generate this independence null. Identity permutations remain valid members of the randomisation distribution and shouldn't be excluded. For three runs, independent identities across all K traits have probability 6^{-K} within one configuration.

Gzipped tarball members require sequential reading from their starts. The reduction therefore extracts the required fields on one pass, since an omitted field requires another complete stream. The default-versus-auxiliary step mismatch at 530M and 750M motivates the auxiliary-two-versus-auxiliary-three contrast, with checkpoint agreement checked separately for both auxiliary runs.

The pipeline lists the 375 checkpoint directories once, applies exponential backoff to rate limits, and caches the listing. Before estimation, it regenerates the common-step table from that cache and compares it with the initial version. This check can detect changes in the recorded step selection, but a cached listing alone can't detect new remote changes without a refreshed comparison.

D.4 COMPUTE

Table 22. Compute budget reported in the planning document for both analysis arms. These figures describe the plan rather than measured costs for every listed operation.

Operation	Budget	Scope
Arm 2 held-out bits per byte	6.0 GPU hours	27 runs, 4,755 items
Calibration simulations N1, N5, N6	1.0 GPU hours	2,000 replicates, six effects
160M checkpoint check	0.5 GPU hours	Adjacent checkpoints
Contingency	2.5 GPU hours	Reserved capacity
Total GPU	10.0 GPU hours	Planned total
Streaming and reduction	18 CPU hours	122.9 GB, one pass
Estimation, bootstrap, permutation	7 CPU hours	Includes 200 planned splits

The primary run used one Kaggle CPU core for Arm 1, reporting 6.0 hours for estimation and null checks. Four sensitivity tasks took another 1.6 hours, including gain calibration and fifty re-splits rather than the 200 budgeted. The exact recipe sweep took thirteen seconds, and the later 160M checkpoint check took 423 seconds of scoring on one T4. Arm 1 performs no forward passes and reports no GPU use.

The transfer procedure streams 25 monolithic tarballs totalling 122.9 GB while keeping peak disk use below 10 GB. Arm 2 performs teacher-forced fp16 scoring with zero-shot contexts and no exemplars. Most items use one context, while WinoGrande uses one per choice. Reported context lengths include 199 tokens for ARC-Challenge and 937 for BoolQ.

The budget assumed two 16 GB cards, but the reported scoring jobs instead received one 46 GB L40S each on a Slurm cluster. Each of 27 jobs finished in under two minutes, with aggregate reported use below one card-hour rather than the budgeted six. Gold-token scoring materialises a float32 log-softmax tensor over a 50,304-token vocabulary. A 30,000-token batch requires approximately 6 GB for that tensor alone.

A configuration rescore fit on a 23 GB card at the planned token budget, while the 16 GB T4 checkpoint check used 6,000 tokens. Changing the budget changes batch grouping, which can affect floating-point results even when the mathematical scoring rule stays fixed. The transport rescore changed hardware and token budget together and shifted inflation by 0.6 percent. CPU handles the reduction and statistical analysis, with model scoring and calibration kernels as the reported exceptions.

D.5 RELEASE PROVENANCE

The null checks examine mechanisms that can affect covariance estimates, but they don't independently establish the provenance of the three released runs. Differences in tokeniser builds, harness revisions, or other training conditions could contribute to the default-versus-auxiliary contrast if those conditions weren't held fixed. The auxiliary-run contrast also relies on documentation that the relevant settings match between its two runs.

The step table and disagreement checks expose some inconsistencies without replacing a release-side audit. Attribution of the observed contrasts to the documented seed changes therefore depends on the release documentation. This limitation applies independently of the estimator's behaviour on simulated or permuted data.

D.6 SENSITIVITY TO GPU ARCHITECTURE

The primary Arm 2 scores use one card type. A separate Turing-versus-Ada comparison on one configuration reports mostly identical item scores, a mean absolute margin difference of 1.0×10^{-3} , and a largest difference of 3.5×10^{-2} . Relative to a margin standard deviation of 0.391, the mean difference has ratio 2.6×10^{-3} .

Eighteen of 4,755 accuracy bits change, or 0.38 percent. Their median absolute margin is 1.3×10^{-3} , and battery accuracy changes from 0.430074 to 0.430494, a difference of 4.2×10^{-4} . That difference

is approximately eight percent of the reported accuracy seed standard deviation. This single comparison demonstrates sensitivity at that scale, but it doesn't establish a universal portability threshold for fp16 kernels.

Arm 1 uses released per-item outputs without recomputing forward passes, so this study doesn't add the same hardware variation to those scores. Shared hardware across Arm 2's 27 runs reduces one source of variation but doesn't independently validate accuracy across architectures.

D.7 CENTRING THROUGH RUN CONTRASTS

Raw cross-half products contain squared configuration means, which the analysis record reports can inflate the ratio by roughly three orders of magnitude. The described implementation projects scores onto an orthonormal basis of mean-zero run contrasts. Each basis row annihilates the all-ones vector, making raw and centred inputs equivalent under projection. Runtime checks test row sums and invariance of $\sum_c T_c$ under a shared configuration shift on both halves, addressing centring errors without testing measurement or exchangeability assumptions.